

The Remains of the Trade: The U.S.–China Trade War and its Aftermath*

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Abstract

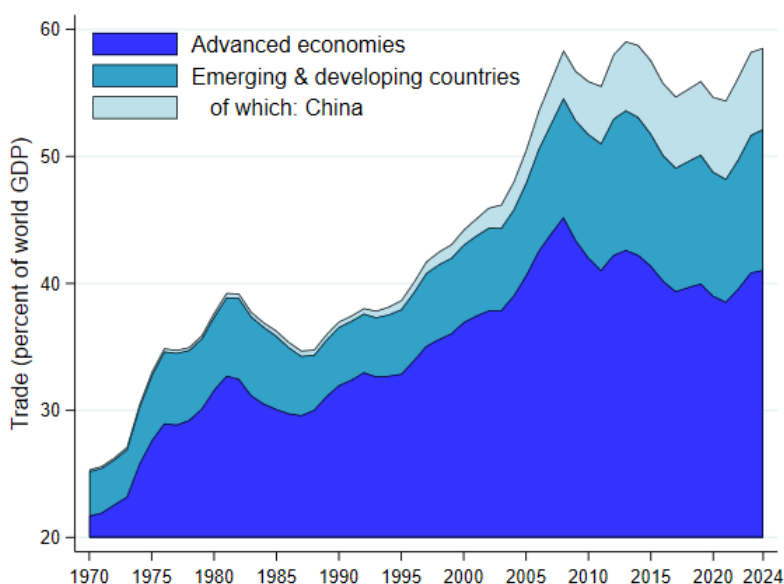
We study how the U.S.–China trade conflict has reshaped global trade patterns with significant spillovers to third-countries. Using recent tariff policy changes and monthly international trade flow data, we document a redirection of Chinese exports away from the U.S. market and toward third-country economies. To interpret the impact of these reallocations on third countries, we introduce a taxonomy that separates four exposure margins – export competition, import competition, consumer opportunities, and sourcing opportunities – and construct theoretically-derived indices that map countries into these channels.

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1 Introduction

Talk of ‘deglobalization’ has become a fixture of policy debates since the mid-2010s, and it has intensified in the wake of the U.S.–China trade war and the COVID-19 shock. Yet, as many authors have emphasized ([Antràs, 2021](#); [Goldberg and Reed, 2023](#)), there is little systematic evidence that, at the global level, international trade has become a declining share of world consumption or income. Figure 1 plots the ratio of world trade (exports plus imports) over world GDP in the period 1970-2024. It is true that after having grown enormously during the late 1980s, 1990s and early 2000s, this ratio has remained flat since the Great Financial Crisis (GFC). Nevertheless, the world is not deglobalizing, it has just ceased to globalize *more*.

Figure 1: World Trade Over World GDP (1970-2024)



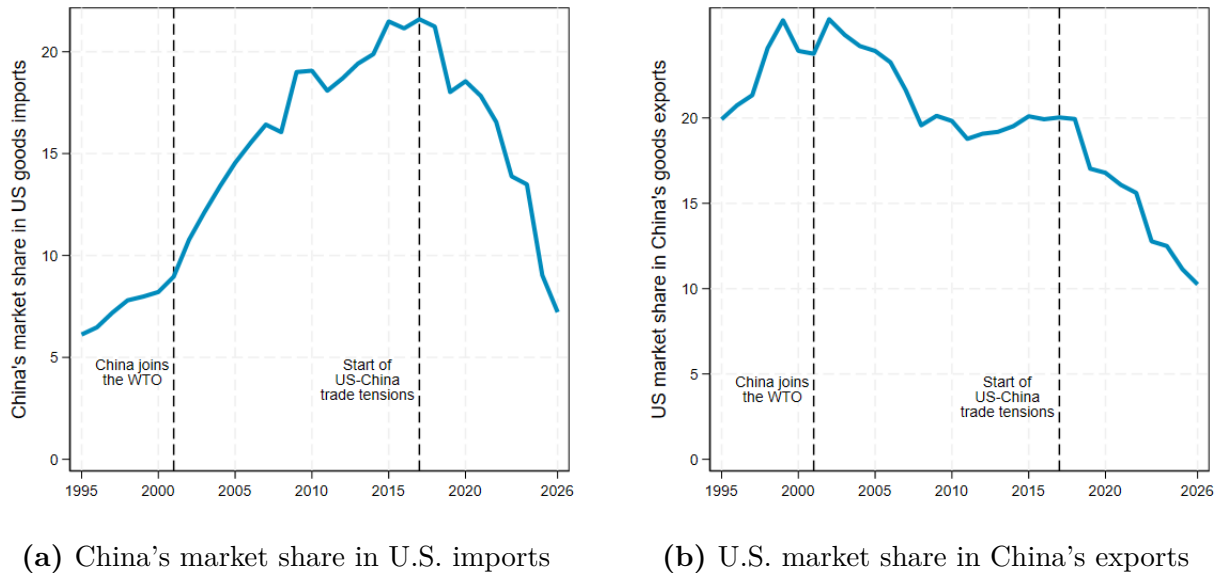
Notes: The figure plots world trade as a percentage of GDP for the period 1970-2024. It decomposes world trade between advanced economies, China, and emerging and developing countries (ex. China). For each region, trade is scaled by world GDP and taken as a backward-looking 3-year moving average.

Sources: World Bank’s World Development Indicators.

At the same time, the recent stability in the world trade-to-GDP ratio masks a notable reshuffling of cross-border linkages. As panel (b) of Figure 1 shows, since the GFC, trade openness has fallen markedly more in advanced economies than in both emerging and developing economies, whether China is included or excluded. Even more notably, direct trade flows between the U.S. and China have tumbled in recent years. As illustrated in the left panel of Figure 2, the share of U.S. imports accounted for by China has fallen drastically from 22% in 2017 to about 8% in 2026, a level not recorded since China entered the WTO in 2001. Similarly, China’s share of exports destined to the U.S. market has fallen from around 20% in 2017 to

about 10% in 2026. Given the sheer size of Chinese exports (\$3.77 trillion in 2025), this entails a redirection of Chinese exports from the U.S. market to other markets comparable in size to the GDP of certain advanced economies such as Chile, Portugal or Greece.

Figure 2: U.S.-China Decoupling



Notes: Panel (a) plots the share of total U.S. goods imports from China, from US Census. Panel (b) plots the share of total China's goods exports to the U.S. The series is from the CEPII-BACI dataset until 2024 and from TDM for 2025 and 2026. For both panels, data for 2026 refer to January and February.

Sources: US Census, CEPII-BACI dataset (Gaulier and Zignago, 2010), and Trade Data Monitor (TDM).

Seen in this light, the key question raised by recent events is not whether globalization is ending, but how trade and production relationships are being reorganized across countries, regions and geopolitical blocs, especially since the onset of the U.S.-China trade war. The defining feature of the post-2018 period is not an indiscriminate retreat from trade, but a ‘Great Reallocation’ (Alfaro and Chor, 2023) of trade flows across partners and products. Much as in Kazuo Ishiguro’s *The Remains of the Day*, the drama here lies less in rupture than in reorganization: continuity in the aggregate, but a profound reshuffling in who trades what with whom, and on what terms.

In this paper, we make three contributions that speak to this post-2018 reorganization. First, we offer a fresh new look at the U.S.–China trade war, shedding new light on how it has developed in recent months – especially since “Liberation Day” – by drawing on up-to-date data to document the latest shifts in trade patterns and trade policy intensity. Second, we zoom in on the effects of the trade war on third countries: we empirically characterize the spillovers of U.S.–China decoupling for third countries (especially emerging markets) rather than merely treating them as residual beneficiaries of diversion. More specifically, we offer

theoretically-motivated diagnostics that separate four exposure margins – export competition, import competition, consumer opportunities, and sourcing opportunities – to organize the evidence. Third, we provide preliminary evidence suggestive that our indices have empirical bite in predicting how the ongoing U.S.-China trade tensions have spilled over to third countries. `textbf[PERHAPS INSTEAD SAY SOMETHING ABOUT OPENING THE BLACK BOX OF THE DIAGNOSTICS]`

Our starting point is the influential work of [Alfaro and Chor \(2023\)](#) documenting systematic changes in U.S. sourcing patterns in recent years, involving a reduction in direct exposure to China and rising shares for alternative suppliers. In section 2, we document some novel aspects of this reallocation with a specific focus on the implications of the U.S.–China trade war in terms of (i) a reallocation of U.S. import sources; (ii) a reorientation of Chinese export destinations; and (iii) broader shifts in global trade patterns. Among other new findings, we find that the initial phases of the U.S.–China trade war saw a relatively slow reallocation of U.S. imports away from China because of the sluggish adjustment associated with products with longer production cycles (cf. [Antràs and Tubdenov, 2025](#)).

More importantly, our analysis also shows that the pace of reallocation has accelerated since the beginning of the second Trump administration and especially since Liberation Day, arguably because that day’s announcements were interpreted as a permanent shock to the U.S.–China trade relationship. Conversely, we find less evidence that other countries facing higher tariffs after Liberation Day saw their exports to the U.S. significantly reduced, likely because those other tariffs were surrounded by reversals and uncertainty. The sharp reduction of Chinese exports to the U.S. was mirrored by sizable increases in Chinese exports to other markets, which generated an intensification of competitive pressures in third markets where China and other exporters compete head-to-head.

Section 3 turns to a more systematic empirical exploration of the implications of the U.S.–China trade war for the countries ‘caught in the middle’. We develop a practical taxonomy to measure third-country exposure to U.S.–China decoupling along four (non-mutually-exclusive) channels: (i) *export competition*, whereby China redirects sales from the U.S. toward third destinations and erodes other exporters’ market shares; (ii) *import competition*, whereby redirected Chinese goods raise import penetration and competitive pressure in domestic markets; (iii) *consumer opportunities*, whereby cheaper imported goods from China reduce prices for consumers in destination markets; and (iv) *sourcing opportunities*, whereby cheaper Chinese intermediates reduce input costs and strengthen downstream competitiveness.

We operationalize these channels with transparent and theoretically-grounded exposure indices, and we also compare them to alternative exposure indices based on export and import similarity measures (cf. [Finger and Kreinin, 1979](#)). We then use these diagnostics to classify countries in terms of their exposure, while allowing for multi-label exposure. **[MORE]**

Our paper relates to several strands of work. First, it speaks to the debate on ‘deglobalization’ and geoeconomic fragmentation, which emphasizes that global integration has not uniformly reversed but is increasingly shaped by security concerns and policy-driven shifts in the organization of trade (Antràs, 2021; Goldberg and Reed, 2023; Aiyar et al., 2023). Second, it connects to the emerging evidence on the ‘great reallocation’ in U.S. sourcing and global supply chains following the U.S.–China tariff war (Alfaro and Chor, 2023, 2025), including work highlighting reallocation, rerouting, and ‘exports in disguise’ through third countries (Malesky et al., 2024; Gopinath et al., 2025; Schulze and Xin, 2025). Third, it complements quantitative assessments of the incidence and welfare consequences of recent U.S. tariff actions and retaliation (Amiti et al., 2019; Fajgelbaum et al., 2020; Cavallo et al., 2021; Flaaen et al., 2020), and uses up-to-date tariff information to situate the post-2018 policy regime (Bown, 2025). **MORE - RECENT LITERATURE ON CHINA 2.0** .

A growing number of papers and analyses—especially in policy circles—point to an increasing impact of the expansion of Chinese exports to other countries, and especially Europe.¹ While the increase of Chinese export globally predates the 2025 tariffs and is the continuation of a secular trend—partly due to weak exchange rate, industrial policy, and subdued domestic demand, consistent with the “vent for surplus” mechanism (Almunia et al., 2021)—the U.S. April 2025 tariffs are a large enough shock to trigger a significant further reallocation of trade flows. Emlinger et al. (2026) study the initial response to the April 2025 U.S. tariffs and do not find robust evidence of a deflection of Chinese exports to Europe until October 2025. Extending the sample to December 2025, Schulte et al. (2026) find that export diversion to Europe has been concentrated in a small tail of highly exposed products, suggesting limited macroeconomic effects. By contrast, Chinese exports seem to have been mostly redirected to emerging and developing countries.

Different from the first China shock, this time Chinese export growth is increasingly concentrated in capital- and technology-intensive sectors (de Soyres et al., 2026). This shift translates into an increasing export similarity between China and advanced economies, while this is not the case for several emerging markets which are less aligned in export structures and may face less competitive pressures. A recent analysis by Chowdhry et al. (2026) support this hypothesis, as increasing Chinese exports displace those of large advanced economies in third markets, especially in more complex products and technologically-intensive industries. However, Chatterjee and Subramanian (2026) argue that developing economies are not insulated from the China shock 2.0. On the opposite, rising Chinese competition—partly still concentrated on traditional, low-skilled goods—does not only imply foregone exports in global markets but also a squeeze for domestic producers. As a result, the China shock 2.0 risks slowing down the

¹ The discussion of the effects of the China shock 2.0 is now at the forefront of the policy debate, as testified by its extensive news coverage—see Financial Times [add REFS], among others.

industrialization of developing countries.

Consistent with our proposed framework, [Aprigliano et al. \(2026\)](#) look at the effects of the China shock 2.0 in the euro area and consider three main margins of adjustment. On the negative side, increasing Chinese exports erode euro area market shares abroad. In addition, through the import competition channels, domestic manufacturing investment declines. However, on the positive side the shock leads to a decline of non-energy goods prices by 1 percent over three years.

The remainder of the paper is organized as follows. Section 2 documents the key empirical patterns of the Great Reallocation associated with U.S.–China decoupling. Section 3 proposes a taxonomy to characterize third-country spillovers. Section 4 concludes.

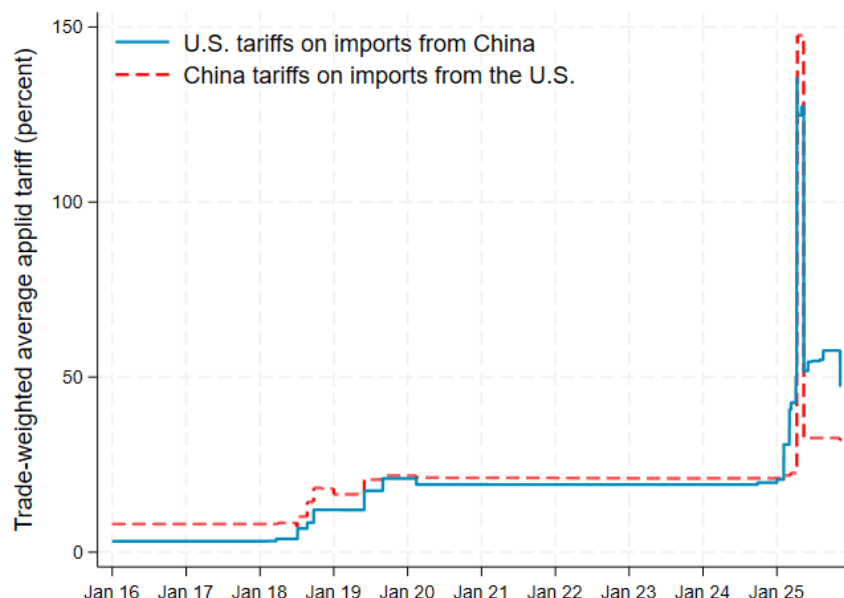
2 The U.S.–China Trade War and the Great Reallocation

The U.S.–China trade war has been one of the most significant economic events in recent times. Early U.S. trade actions included Section 201 safeguard tariffs on solar products and washing machines and Section 232 tariffs on steel and aluminum, but the dispute quickly broadened into large-scale, China-specific Section 301 tariffs. Over 2018–19, successive Section 301 rounds expanded from targeted product lists to cover most bilateral trade, pushing the trade-weighted average U.S. tariff on imports from China from a low pre-war baseline into double digits (see Figure 3, based on [Bown, 2025](#)). China followed suit and retaliated in a commensurate manner. The January 2020 “Phase One” agreement halted further escalation and reduced some tariff rates, but most duties remained in place and bilateral trade barriers stayed far above pre-2018 levels, with a trade-weighted average of roughly 20% in both countries.

From 2021 through 2024, this tariff structure was largely maintained with only selective and sector-specific adjustments. In 2025, however, the conflict entered a much more volatile phase. A new wave of tariff actions associated with President Trump’s April 2, 2025 “Liberation Day” tariffs and related executive orders pushed effective tariff rates sharply higher (including brief peaks at triple-digit levels) before partial rollbacks, as seen in Figure 3 ([Bown, 2025](#)). On May 12, 2025, a joint U.S.–China statement issued after talks in Geneva committed the United States to remove many of the duties imposed in early April, but as of November of 2025, the trade-weighted average U.S. tariff on imports from China remained at 47.5% (see Figure 3, and [Bown, 2025](#)).² The scale and framing of the 2025 escalation were widely read as evidence that the U.S.–China relationship is being redefined on geopolitical grounds, making the shock appear long-lasting (or even permanent) rather than transitory. With that interpretation in mind, we now document how international trade flows have adjusted.

²The ratio of *collected* customs duties to U.S. imports from China has settled at a slightly lower level of around 40%, as reported by Mike Waugh’s “Trade War Tracker” <https://www.tradewartracker.com/>.

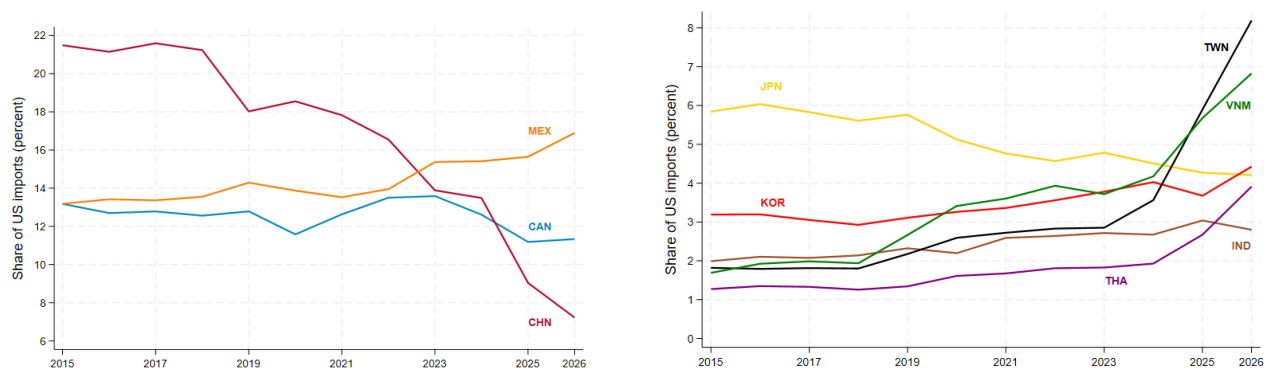
Figure 3: The U.S.–China Trade War: Trade-Weighted Average Tariffs 2016-2025



Notes: The series plot trade-weighted average applied tariffs. The solid blue line is the U.S. tariff on imports from China; the dashed red line is China's tariff on imports from the United States. The PIIE series begins in 2018; 2016-2017 are shown as a flat extension at the January 2018 level for display. The series excludes antidumping/countervailing duties.

Source: Bown (2025).

Figure 4: The Great Reallocation: U.S. import shares (2017-2025)



(a) U.S. import shares for large exporters

(b) U.S. import shares for some smaller exporters

Notes: Shares are calculated as each partner's goods imports divided by total for all economies. Partner shares computed for China, Mexico, Canada, Japan, Vietnam, Taiwan Province of China, South Korea, India, and Thailand. Data for 2026 refer to the period January-April.

Sources: U.S. Census Bureau (<https://www.census.gov/foreign-trade/balance/index.html>).

The Great Reallocation

Against that backdrop, the central adjustment has been a reorganization of trade relationships – a ‘Great Reallocation’ of U.S. sourcing patterns – with broader implications for third countries. Figure 4 summarizes the evolution of U.S. import market shares across major partners from 2015 April to 2026 (the most recent month for which data is available). The dominant fact is an initially gradual but recently more pronounced decline in China’s share. Overall, the Chinese share in U.S. imports fell from around 22 percent in 2017 to around 8 percent in 2026. This decline was mirrored by gains for a set of alternative suppliers, most prominently Vietnam, Mexico and Taiwan. Importantly, the timing of these gains is uneven: Vietnam’s rise is relatively gradual, whereas Taiwan, Thailand and Mexico’s gains are much more concentrated in the later years of the reallocation (2024–2026).

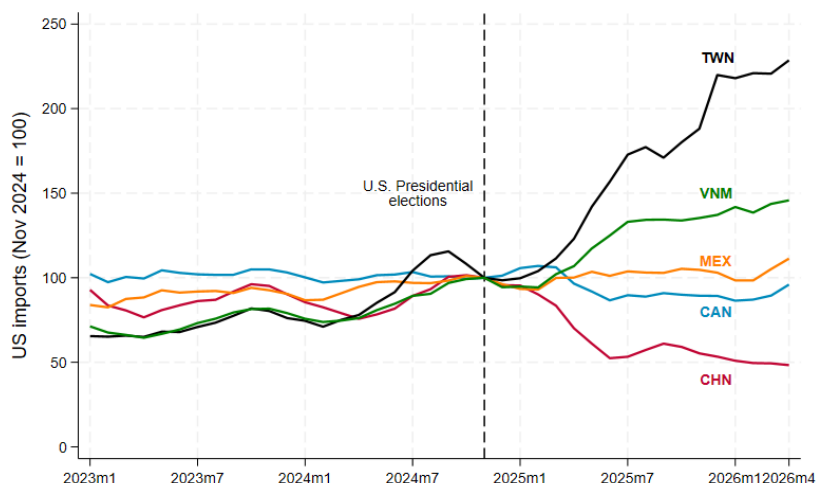
Much has been written about the slow initial decline in U.S. imports from China. The COVID-19 pandemic was certainly a contributor: although imports fell at the onset of the pandemic, they surged back within months as consumers shifted toward goods and e-commerce, helping propel a strong rebound in U.S. goods imports from China. Another key reason why the early tariff shock did not immediately translate into a one-for-one collapse of Chinese exports to the United States is that supply chains feature meaningful adjustment frictions. Antràs (2021) emphasized the importance of sunk costs in generating hysteresis in importing given the large policy uncertainty in the early 2020s, while Alfaro and Chor (2025) have recently shown that the reallocation away from China was slower in contract-intensive and relationship-sticky goods. In Appendix A.1, we provide evidence for a novel channel contributing to the gradual nature of the U.S.–China decoupling. More specifically, building on a measure of the duration of production developed by Antràs and Tubdenov (2025) – the so-called average period of production or APP – we show that reallocation of Chinese exports away from the U.S. was faster in products with a shorter APP , while products with longer production cycles adjusted more slowly.

Trump 2.0 and Liberation Day

Returning to the more drastic reallocation observed since the start of President Trump’s second administration, Figure 5 uses monthly data to zoom in on the impact on U.S. imports for selected trading partners during 2025. The figure illustrates the remarkable drop in U.S. imports from China coupled with a reallocation of those import flows to Taiwan, Vietnam and to a lesser extent Mexico. Imports from Canada fell during this period. Interestingly, with the exception of China, there appears to be a *positive* relationship between import growth and the tariffs imposed on these partners on Liberation Day, since Mexico and Canada were imposed a 10 percent tariff, while Taiwan Province of China and Vietnam were given tariff rates of 32

and 46 percent, respectively. Beyond these five trading partners, there does not seem to exist a strong systematic relationship between changes in U.S. imports and the tariff rates faced by partner countries. Figure 6, inspired by Figure 9 in Alfaro and Chor (2025), may be interpreted as suggesting that countries facing higher new tariff rates tended, on average, to see weaker growth in U.S. imports. However, this pattern is not robust: it is largely driven by China, and it fades when the relationship is estimated via ordinary least squares (rather than when weighting observations by import value, as in Alfaro and Chor, 2025).³

Figure 5: The Great Reallocation during Trump’s Second Administration



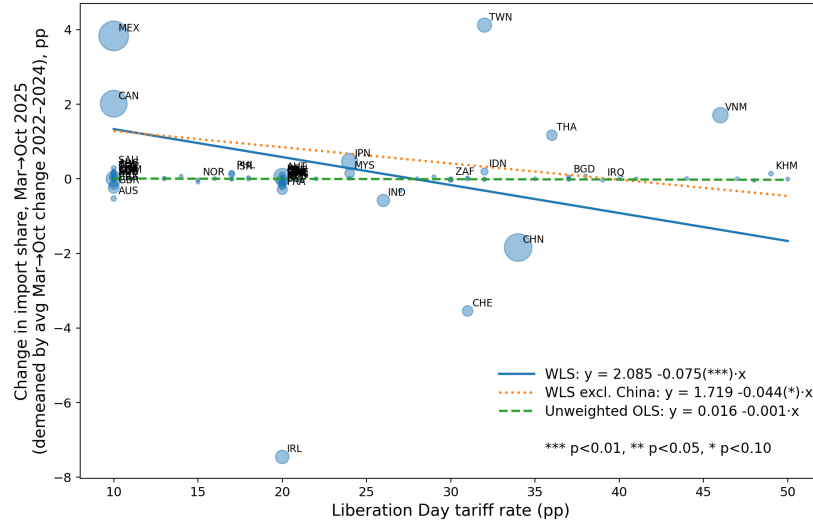
Notes: Series show U.S. goods imports from each partner, indexed to Nov. 2024 = 100. Values are a 3-month backward moving average (month t equals the average of $t-2$, $t-1$, and t). Vertical dotted line marks Nov. 2024.

Source: U.S. Census Bureau (<https://www.census.gov/foreign-trade/balance/index.html>).

Liberation Day also led to a remarkable reorientation of Chinese exports. Figure 7 tracks China’s exports to the United States relative to its exports to other destinations, normalized to November 2024. After the November 2024 election, Chinese exports to the U.S. declined sharply, while exports to Europe and to other Asian economies rose, suggesting a rapid diversion of Chinese sales away from the U.S. market and toward alternative destinations. This prospect has heightened anxiety in the recipient markets: in Europe, policymakers explicitly warned about trade diversion and moved to strengthen import-surveillance capacity to detect and respond to sudden surges, amid concerns about being “flooded” by redirected Chinese goods. In parts of Asia, similar fears about displacement and import surges have been reflected in a more defensive stance – e.g., new or extended anti-dumping actions against specific Chinese products

³The legend in Figure 6 includes the regression results. The weighted least squares coefficient excluding China is negative but much smaller and significant only at the 10 percent level. The ordinary least squares coefficient (including China) is essentially 0.

Figure 6: Liberation Day and Sources of U.S. Imports



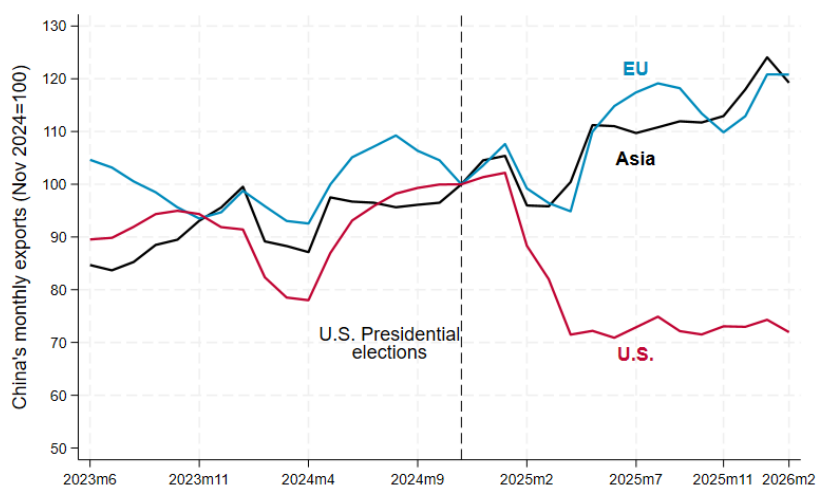
Notes: Each dot is a U.S. import partner. The y-axis is the change in the partner’s share of total U.S. imports from March to October 2025, demeaned by the partner’s average March-to-October change over 2022-2024. Dot size is proportional to the partner’s 2024 total imports. Lines are linear fits: import-weighted (weights = 2024 imports), import-weighted excluding China, and unweighted OLS; significance stars denote $p < 0.10$, * $p < 0.05$, ** $p < 0.01$.

Source: U.S. Census Bureau import totals by partner; Liberation Day country tariff rates from the April 2, 2025 Executive Order, Annex I.

– alongside broader public concern about the competitive pressure created by low-priced Chinese exports being redirected into regional markets. **[[ADD REFERENCES; CHINA 2.0]]**

The reorientation of U.S. imports and Chinese exports is mirrored in broader shifts in global trade patterns. Figure 8 plots the change in bilateral import shares (in panel (a)) and export market shares (in panel (b)) for various countries and regions when comparing the year after the U.S. election (Nov. 2024 – Feb. 2026) to the previous period Jun. 2023 – Oct. 2024. At one level, the heat map reiterates the same basic narrative in Figures 5 and 7: the share of China’s exports flowing to the U.S. market contracts sharply (-3.3 p.p.), but it increases in Asia (+2.4 p.p.) and the rest of the world (+0.9 p.p.). On the flip side, the share of U.S. imports originating in China falls by 3.5 p.p., with large increases from Asia (+1.5 p.p.) and the rest of the world. But Figure 8 also makes clear that the post-election reallocation is not confined to the U.S.–China dyad: there are sizeable market-share shifts in third markets that do not directly involve either country. For example, the America bloc (Canada–Mexico) reoriented their exports towards China, the EU market, and rest-of-the-world destinations (+0.2 p.p.). At the same time, countries in the European Union recorded gains in import shares across China (+0.6 p.p.), developing Asia (+0.6 p.p.), Canada and Mexico (+0.2 p.p.), while reducing imports from the rest of the world region (-1.6 p.p.).

Figure 7: The Reallocation of Chinese Exports during Trump’s Second Administration



Notes: Exports are 3-month backward-looking moving averages. The vertical line is drawn in correspondence with the November 2024 U.S. presidential election. The Asia aggregate includes Bangladesh, Cambodia, Hong Kong SAR, India, Indonesia, Malaysia, Pakistan, the Philippines, Singapore, Sri Lanka, Taiwan Province of China, Thailand, and Vietnam. EU = European Union.

Sources: Trade Data Monitor.

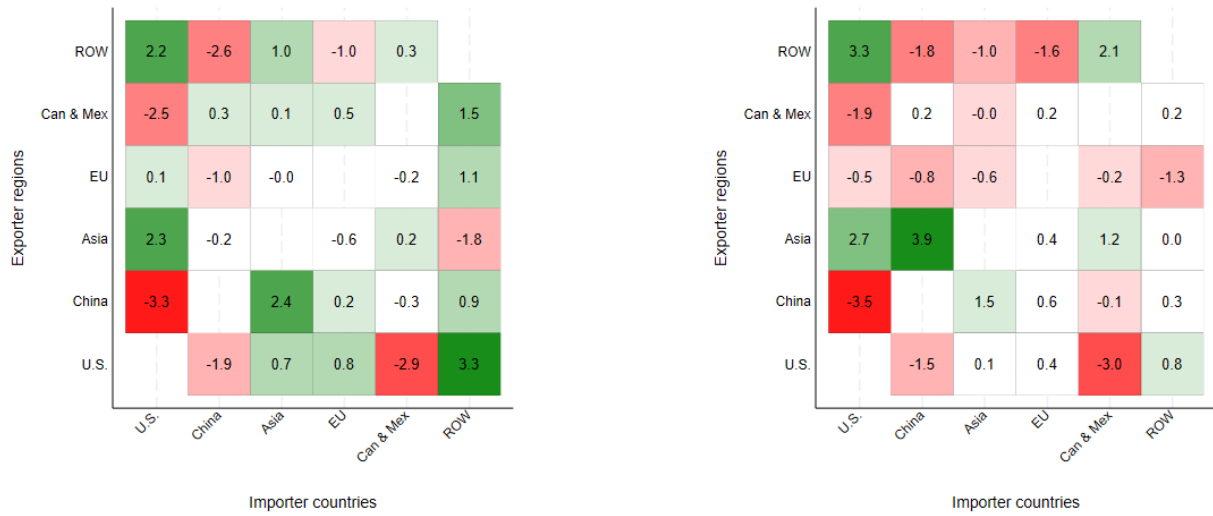
Taking Stock

Taken together, Figures 5–8 suggest that Liberation Day may have been interpreted less as a temporary escalation and more as a permanent shock to the U.S.–China trade relationship – an inflection point toward deeper, sustained decoupling. While other countries were also hit with higher U.S. tariffs, those measures were surrounded by reversals and uncertainty, so it is unsurprising that we see no clean, systematic trade response. By contrast, the sharp curtailment of China’s ability to sell into the U.S. market generated sizable spillovers elsewhere: Chinese exports were redirected into other destinations, and competitive pressures intensified in third markets where China and other exporters meet head-to-head. We next turn to a more systematic analysis of these spillover effects to third markets.

3 Caught in the Middle: Third-Country Spillovers from the U.S.–China Tariff War

When two giants fight, smaller economies do not simply “sit out” the conflict. The new phase of the U.S.–China tariff war has curtailed China’s sales into the U.S. market and, in turn, altered the competitive environment faced by firms around the world. This section examines these third-country spillovers systematically.

Figure 8: A Worldwide Reallocation during Trump's Second Administration



Notes: The two charts report the change in export market shares (panel a) and import market shares (panel b) in percentage points, computed around the 2024 U.S. election, comparing Nov. 2024–Feb. 2026 to Jun. 2023–Oct. 2024. The Asia aggregate includes Bangladesh, Cambodia, Hong Kong SAR, India, Indonesia, Malaysia, Pakistan, the Philippines, Singapore, Sri Lanka, Taiwan Province of China, Thailand, and Vietnam. EU = European Union. ROW = Rest of the world. Changes in market shares sum to zero by row in panel (a) and by column in panel (b).

Sources: Trade Data Monitor.

A natural starting point is trade diversion in the narrow sense: third countries may gain U.S. market share as Chinese suppliers are displaced, or they may gain market share in China if U.S. sellers are displaced. The patterns since Liberation Day, however, offer limited support for this narrow diversion narrative. The cross-partner relationship between U.S. import growth and the new tariff schedule appears weak (Figure 6), and the relatively small scale of U.S. goods exports to China (compared with U.S. imports from China) implies that, for many countries, direct bilateral exposure to China through that channel is second-order. These observations motivate a different perspective in which the main action occurs in markets *other* than the U.S. itself.

Our emphasis is therefore on spillovers generated by China’s impaired access to the U.S. market but realized in third markets. We decompose exposure into four conceptually distinct margins. First, *export competition*: as Chinese exporters redirect shipments, they displace other suppliers in shared destination markets. Second, *import competition*: redirected Chinese goods raise import penetration and competitive pressure in domestic markets. Third, *consumer opportunities*: those same redirected Chinese goods will tend to be sold at lower prices than the purchases they displace, which will reduce the price level for consumers in those import markets, and perhaps also increase the variety of goods available to them. Fourth, *sourcing opportunities*: lower-priced (or more readily available) Chinese intermediates can reduce input costs and expand downstream activity. A special case of this latter channel reflects a *rerouting opportunity*, and captures how cheaper Chinese imports enhance exports specifically to the U.S. market, which can be interpreted as Chinese firms circumventing U.S. barriers by shifting assembly, routing trade through intermediaries, or relocating stages of production to countries with better market access to the United States.

We next operationalize this taxonomy with tractable diagnostics and then use them to characterize which countries are most exposed along each margin. Before we dive into the specifics of our indices, we offer some broad observations applying to all these indices.

3.1 General Considerations

Our goal is to develop measures of the extent to which countries are exposed to an increase in Chinese exports to countries other than the U.S., resulting from the constrained ability of these same Chinese exporters to sell to the United States. At a very general level, it is clear that countries are more likely to face export competition from China if their export patterns are *similar* to those of China. At the same time, the other channels of exposure should be more salient when a country’s reliance on imports from China is *large* and *similar* in structure to that of the U.S., since this would indicate that U.S. policies would divert relatively more Chinese exports toward that exposed country.

A common approach to capture similarity in exports and imports follows from the influential

work of [Finger and Kreinin \(1979\)](#). Let us review the basics of their index for the case of exports. Suppose that one is interested in assessing how similar is the export structure of two countries, i and j . By export structure, we mean a vector $\mathbf{X}_o = \{X_{o11}, \dots, X_{odp}\}$ where X_{odp} refers to (origin) country o 's exports of product p in (destination) market d . [Finger and Kreinin \(1979\)](#) proposed the following similarity index between the exports of country i and j in destination market d :

$$ESI_{ijd} = \sum_{p \in \mathcal{P}} \min \left(\frac{X_{idp}}{\sum_{p' \in \mathcal{P}} X_{idp'}}, \frac{X_{jdp}}{\sum_{p' \in \mathcal{P}} X_{jdp'}} \right) \times 100. \quad (1)$$

To see how this measures captures export similarity, notice that if the countries i and j feature the same *share* of exports in all goods p , then $ESI_{ijd} = 100$ because the two countries are 100 percent similar in their export structure in the destination market d . Conversely, if there is *no* product p for which *both* countries feature positive exports, then their similarity is zero.⁴

The [Finger and Kreinin \(1979\)](#) index is an insightful index of export similarity, but it is much less clear that it constitutes a good index of export competition pressures. As pointed out by [Jenkins \(2008\)](#), the fact that the index ESI_{ijd} in equation (1) is symmetric is problematic: when interpreted as capturing competitive pressures, it would imply that China is as impacted by competition from Honduras as Honduras is impacted by competition from China. Furthermore, the index is also symmetric across products and thus does not take into account that competitive pressures might be felt more intensively in some products than others.

With that in mind, below we will develop alternative indices of exposure that are theoretically grounded, and we will compare them with the corresponding similarity measures. More specifically, in Appendix A.2, we develop an Armington model with product-level CES demand: in each destination market, expenditure on product p is allocated across exporter varieties according to a constant elasticity. A China-specific U.S. policy shock reduces Chinese exports to the U.S. and, via an upward sloping Chinese export supply function, leads to lower prices in all other destinations. The model then traces the first-order implications of this shock on the exports of a given country j to all destination markets other than the U.S. and China.

We construct all our exposure measures based on exports and imports data. We use bilateral trade data at the product level sourced from Trade Data Monitor (TDM), which compiles monthly trade data from government offices and national agencies. Data are available with a lag of just a few months and cover monthly goods trade (imports and exports) from 104 reporting economies to 195 partner economies, and are disaggregated at the six-digit Harmonized System (HS-6) level, covering 7,352 unique products. We construct a balanced panel of trade flows (in nominal USD) between 195 countries using imports from country A to country B to measure

⁴As shown by [Sun and Ng \(2000\)](#), the [Finger and Kreinin \(1979\)](#) index has good axiomatic properties, in the sense that it is invariant with respect to proportional sub-classifications.

exports from B to A, when the latter is not available because B is not a reporting country.

3.2 Diagnosing Export Exposure

With this background in mind, we begin by proposing indices capturing the extent to which third countries may face increased export competition from China in countries other than the US (and China) as a result of the inability of Chinese exporters to sell to the United States. We begin with the standard approach of producing measures of export similarity, but then propose our theoretically-grounded index of export competition.

3.2.1 Export Similarity

As argued before, export competition from China might be expected to be fiercer for countries with an export structure relatively more similar to that of China. To measure the export similarity between a given country j and China, we first compute the following variant of the [Finger and Kreinin \(1979\)](#) export similarity index (ESI):

$$ESI_j = 100 \times \sum_{d \in \mathcal{W}_-} \sum_{p \in \mathcal{P}} \min \left(\frac{X_{jdp}}{X_j}, \frac{X_{cdp}}{X_c} \right), \quad (2)$$

where c denotes China, X_{jdp} are country j 's exports of product p in destination d (in some baseline year), $\mathcal{P}$ is the set of all products, $\mathcal{W}_-$ is the set of all countries in the world ($\mathcal{W}$) excluding the U.S. and China, and where X_j and X_c are aggregate exports of j and China in all countries other than U.S. and China (in the baseline year). To avoid anticipation effects in the run-up of the U.S. presidential election, we choose 2023 as our baseline year.⁵

In robustness tests, we also compute coarser indices of export similarity that only consider similarity in the types of products exported (regardless of where they are sold) and in the destination of sales (regardless of which goods are sold in those destinations). The product-level index, when computed in some baseline year is given by

$$ESI_j^{prod} = 100 \times \sum_{p \in \mathcal{P}} \min \left(\frac{\sum_{d \in \mathcal{W}_-} X_{jdp}}{X_j}, \frac{\sum_{d \in \mathcal{W}_-} X_{cdp}}{X_c} \right), \quad (4)$$

and it may capture better than equation (2) situations in which the shock (e.g., Liberation

⁵We have also computed a variant of the export similarity index that aggregates across destinations as follows:

$$ESI_j = 100 \times \sum_{d \in \mathcal{W}_-} \frac{X_{jd}}{X_j} \sum_{p \in \mathcal{P}} \min \left(\frac{X_{jdp}}{X_{jd}}, \frac{X_{cdp}}{X_{cd}} \right), \quad (3)$$

which amounts to an export-weighted average of the destination-specific export similarity indices with China. The correlation.

Day) leads to a disproportionate increase in Chinese exports to destinations in which it sold relatively less in the baseline year. On the other hand, the coarser similarity index based on the destination of sales is given by

$$ESI_j^{dest} = 100 \times \sum_{d \in \mathcal{W}_-} \min \left(\frac{\sum_{p \in \mathcal{P}} X_{jdp}}{X_j}, \frac{\sum_{p \in \mathcal{P}} X_{cdp}}{X_c} \right), \quad (5)$$

and it may better capture concerns about China disproportionately increasing the sales of *new* types of goods in markets in which both j and China had sizeable market shares in the baseline year.

Table 1 lists the 20 self-governed economies that, according to export similarity at the product and destination levels (see equation (2)), face the highest export exposure to China.⁶ Most of these most exposed economies are in Asia, although the table also includes a number of European economies, such as Germany or Italy.

Table 1: Top 20 economies by Export Similarity (ESI_j)

Rank	Country	ESI_j	Rank	Country	ESI_j
1	Vietnam	29.25	11	Philippines	16.50
2	South Korea	25.70	12	Czechia	16.17
3	Japan	24.61	13	France	16.01
4	Taiwan	23.67	14	United Kingdom	15.62
5	Thailand	23.58	15	Singapore	15.34
6	India	20.60	16	Poland	14.72
7	Italy	20.25	17	Turkey	14.64
8	Germany	20.16	18	Mexico	14.34
9	Malaysia	20.02	19	Netherlands	14.25
10	United States	17.64	20	Indonesia	13.90

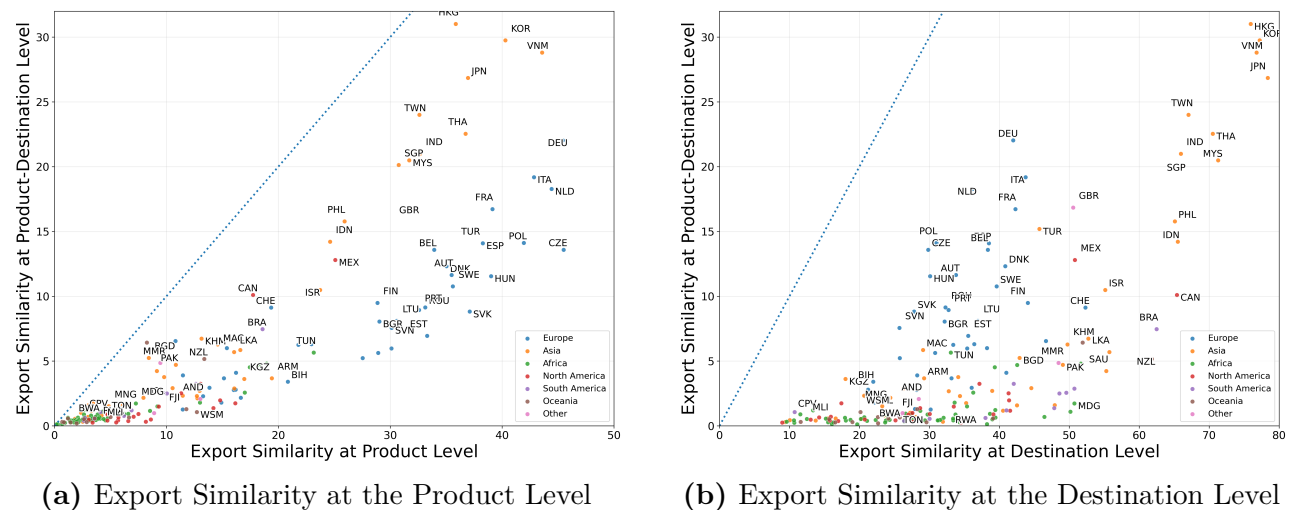
Figure 9 plots our benchmark export similarity measure against the two previously mentioned coarser measures that allow for more flexible channels of export competition. Panel (a) plots export similarity when computed at the product level. The correlation between the two indices is very high (0.87) but, at the product level, export similarity with China is always larger. The figure further indicates that European countries tend to be much more impacted by this measure, reflecting the fact that these countries export the same products as China does, but not necessarily to the same destinations. This further implies that if Chinese exporters were to

⁶We exclude Hong Kong SAR, which has the highest export similarity with China at 31.01. In later tables we also exclude Macao SAR and other non-self-governed territories. Values for all countries and territories are provided on our support website. [ADD LINK IN DUE COURSE]

reorient their sales to these destinations currently favored by European firms, the shock could be much more intensively felt in European countries.

Panel (b) is analogous but plots export similarity when computed at the destination level. This captures threats stemming from China ramping up their exports of new or low-selling products in the destinations in which the potentially impacted countries concentrate their sales in the baseline year. European countries seem to be more immune from this shock, but many Asian economies – as well as Canada and Brazil – would face much fiercer enhanced export competition on this account.

Figure 9: Alternative Measures of Export Similarity



3.2.2 An Export Competition Index

We next turn to our theoretically-grounded index of export competition. We omit all the technical details in the main text, but the expressions below are derived from an Armington model in Appendix A.2. The percentage response of country j 's exports is shaped by three key empirical objects.

a) Size of product-level shock A first consideration is the size of the U.S.-China tariff shock for different products p . More specifically, we seek to measure by how much will (free-on-board) Chinese prices fall in third markets as a result of the tariffs set by the U.S. on China, and we denote that Chinese price decline by Φ_p . For uniform U.S. tariff changes and uniform demand and supply elasticities across sectors, the relative size of the shock across sectors turns out to be governed solely by the share of overall Chinese production of good p that was shipped to the

U.S. (relative to other destinations) in the baseline year, or

$$\phi_p = \frac{X_{cup}}{\sum_{d \in \mathcal{W} - \{c\}} X_{cdp}}. \quad (6)$$

Intuitively, the larger is the share of Chinese output that was sold in the U.S., the larger will be the amount of exports that will need to be redirected to other markets, and with an upward sloping Chinese export supply, the larger will be the necessary price decline in other markets.

Table 2: Top 20 HS6 Products by ϕ_p

Rank	HS6	Description	ϕ_p	Rank	HS6	Description	ϕ_p
1	030356	Fish; frozen, cobia...	1.000	11	842860	Teleferics, chair-l...	0.844
2	030242	Fish; fresh or chil...	0.973	12	030615	Crustaceans; frozen...	0.839
3	720450	Ferrous products; r...	0.950	13	300449	Medicaments; contai...	0.804
4	030563	Fish; salted or in ...	0.949	14	121292	Locust beans (carob...	0.770
5	291431	Ketones; aromatic, ...	0.948	15	292640	Nitrile-function co...	0.758
6	711031	Metals; rhodium, un...	0.947	16	030531	Fish fillets; dried...	0.733
7	293080	Organo-sulphur comp...	0.943	17	902229	Apparatus based on ...	0.724
8	130214	Vegetable saps and ...	0.913	18	900669	Photographic flashl...	0.711
9	030462	Fish fillets; froze...	0.875	19	282540	Nickel oxides and h...	0.691
10	090112	Coffee; decaffeinat...	0.861	20	680210	Tiles, cubes and si...	0.684

Note: HS codes with missing elasticity estimates or tariff data are excluded.

Table 2 reports the twenty HS6 products with the highest values of ϕ_p in 2023, excluding special classification codes.⁷ The products are concentrated in a relatively narrow set of primary commodities (e.g., seafood and cobalt ores), specialized chemical inputs, and selected high-value metals such as rhodium and beryllium. High values of ϕ_p indicate that, for Chinese exporters of these goods, the U.S. market represents a particularly important destination. Exports of these products from China are therefore disproportionately oriented toward the United States.⁸

As the model in Appendix A.2 shows, the size of the product-level tariff shocks is not only shaped by shares of Chinese exports going to the U.S., but also by the tariff changes and the demand and supply elasticities in those products. More specifically, the implied decline in the Chinese (f.o.b) price of good p in other destinations is given by

$$\Phi_p = -d \log p_{cp} = \phi_{cup} \times \frac{\varepsilon_{cup}}{\bar{\varepsilon}_{cp} + \eta_{cp}} \times \Delta \log (1 + \tau_{cup}). \quad (7)$$

⁷Table C.1 reports the twenty HS6 products with the highest values of Φ_p , and decomposes it into the components of (7).

⁸Importantly, these results should *not* be interpreted as evidence that China is a critical supplier of these goods for the United States. A high ϕ_p reflects the dependence of Chinese exporters on the U.S. market, not the dependence of the U.S. on China as a source country.

In this expression, ε_{cup} is the U.S. demand elasticity for Chinese product p , $\bar{\varepsilon}_{cp}$ is the average import-demand elasticity faced by China across export markets, η_{cp} is the elasticity of supply of Chinese exports of good p , and $\Delta \log(1 + \tau_{cup})$ is the change in US tariffs on imports from China of good p .

We construct a proxy for Φ_p based on ϕ_p , demand and supply elasticities from [Soderbery \(2018\)](#), and tariff changes from the Global Tariff Database in [Rodríguez-Clare et al. \(2025\)](#) (based on the methodology developed in [Teti, 2024](#)). The details are in Appendix B. The correlation between Φ_p and ϕ_p is very high (0.822), suggesting that most of the variation in the shock is coming from the transparent and observable Chinese share of exports destined to the U.S., rather than the more debatable elasticity estimates. Indeed, a variance decomposition of equation (7) indicates that more than 90% of the variation in Φ_p is driven by ϕ_p , with elasticities accounting for around 7% and tariff changes for the remaining 3%.⁹

b) Incidence of product-level shock in destination market d The second key variable shaping exposure is the extent to which Chinese exports of good p diverted to a given destination d will crowd out other sellers of good p in that market. Our Armington-CES model indicates that this force can be approximated (up to a first order) by the share of expenditure in country d that is accounted for by Chinese exports, or

$$\chi_{cdp} = \frac{X_{cdp}}{\sum_{o \in \mathcal{W}} X_{odp}}. \quad (8)$$

The term χ_{cdp} captures the incidence of the shock in destination market d . Intuitively, a fall in the price of Chinese exports of product p generates larger competitive pressure in markets where China is already an important supplier of that product. If China accounts for only a negligible share of purchases of product p in destination d , then even a sizable fall in the Chinese price has little first-order effect on the revenues of other exporters in that market. By contrast, when χ_{cdp} is large, Chinese exporters are already close competitors in that product–destination cell, so a reduction in their relative price produces a larger displacement of other suppliers.

c) Relevance of the shock for origin country o Finally, the effect of this destination–product shock on the origin country o depends on how important that same product–destination

⁹We compute tariff changes from January 2025 (before Trump’s second inauguration) to September 2025. Later, we will consider robustness to a ‘long’ variant with export shares computed from 2018 data and tariff differences from January 2018 to September 2025. This has no bearing for the decomposition above: ϕ_p continues to account for more than 90% of the variation in Φ_p , while the contribution of tariff changes actually drops to 2%.

cell is for country o 's own exports. For a given destination d , define

$$s_{odp} = \frac{X_{odp}}{X_{od}}, \quad (9)$$

where $X_{od} = \sum_{p \in \mathcal{P}} X_{odp}$. Thus, s_{odp} is the share of country o 's exports to destination d accounted for by product p . This term captures the relevance of the shock for the origin country: even if Chinese competition in product p is intense in destination d , country o is exposed only to the extent that it also sells product p in that destination. Conversely, if product p represents a large share of o 's exports to d , then displacement in that product–destination cell has a larger effect on o 's exports to that market.

The Export Competition Index Finally, aggregating across destinations – weighting each destination by its importance in country j 's overall export basket – the exposure of country j 's overall exports to the U.S.–China trade shock is given by

$$ECI_j = 100 \times \sum_{d \in \mathcal{W}_-} \frac{X_{jd}}{X_j} \sum_{p \in \mathcal{P}} (\sigma_{jp} - 1) \times s_{jdp} \times \chi_{cpd} \times \Phi_p \quad (10)$$

Here $\mathcal{W}_-$ is the set of all countries in the world excluding the U.S. and China. The index therefore combines three forces: the size of the China price shock in product p , Φ_p ; the intensity of Chinese competition in destination market d , χ_{cpd} ; and the importance of that product for origin country j 's exports to destination d in country j 's total exports.

The term $\sigma_{dp} - 1$ in (10) appears because the relevant mechanism is substitution across country-of-origin varieties within a destination–product market. A U.S. tariff reduces Chinese export demand in the U.S.; with upward-sloping Chinese export supply, this lowers the Chinese f.o.b. price and therefore the delivered price of Chinese varieties in third markets. The revenue loss for other exporters then depends on how strongly buyers in destination d substitute toward the now-cheaper Chinese variety. When the elasticity of substitution is high, a given fall in China's relative price induces a larger reallocation of expenditure away from other origins and toward China. When demand is less elastic, the same price change generates less displacement. This is why the model weights the China price shock by the demand elasticity term.

There are three main differences between the export competition index in equation (10) and the Finger-Kreinin similarity index in equation (2). First, although our index includes information on both country j 's export shares s_{jdp} and China's market shares χ_{cpd} , these shares are conceptually different from each other. In particular, for China what matters is the relative importance of Chinese exports in total absorption of good p in destination d , rather than how important those Chinese exports are relative to overall Chinese exports (including exports of other goods and to other destinations). Second, the model indicates that exposure is

proportional to the product of these shares χ_{cdp} and s_{jdp} rather than the minimum of the two, as in the similarity measure (2). Third, equation (10) also captures the fact that the shock may be more disproportionately felt in some sectors or products than others, and it shows that this heterogeneity can be proxied by the share of Chinese exports of good p that were destined to the U.S. market relative to other markets in the baseline year.

We compute our export competition index with the same data used to compute the Finger-Kreinin export similarity index in (2). Because we do not have data on domestic production disaggregated at the HS-6 product level, the denominator of χ_{cdp} in equation (8) does not include country d 's domestic sales of product p .¹⁰

Table 3 reports the top-20 economies by the Export Competition Index with China (ECI_j). Remember that the figures in the table correspond to the estimated percentage declines of these countries' exports due to increased Chinese competition. The ranking is headed by a set of relatively small and/or specialized economies, but some notable medium-sized exporters, such as Vietnam or Hungary, emerge. Taiwan is also part of the top 10 most affected countries. Overall, Table 3 points to exposure that is concentrated in smaller, export-specialized economies in Asia and a handful of small island or low-income countries.

Table 3: Top 20 economies by Export Competition Index (ECI_j)

Rank	Country	ECI_j	Rank	Country	ECI_j
1	Armenia	2.54	11	Myanmar	1.13
2	Vietnam	1.81	12	Lesotho	1.13
3	Kyrgyzstan	1.67	13	Laos	1.06
4	Andorra	1.54	14	Czechia	1.06
5	Mongolia	1.52	15	Mexico	1.04
6	Cambodia	1.47	16	Bangladesh	1.03
7	Mali	1.36	17	Poland	1.00
8	Hungary	1.29	18	Thailand	0.98
9	St. Kitts & Nevis	1.23	19	Sri Lanka	0.97
10	Romania	1.15	20	Turkey	0.93

Relative to the export similarity in Table 1, the composition shifts markedly away from large diversified manufacturing exporters (e.g., South Korea, Japan, Germany, and several large European economies) and toward smaller economies. There is some overlap: Vietnam and Taiwan rank near the top in both tables, and a few additional economies (Thailand, the

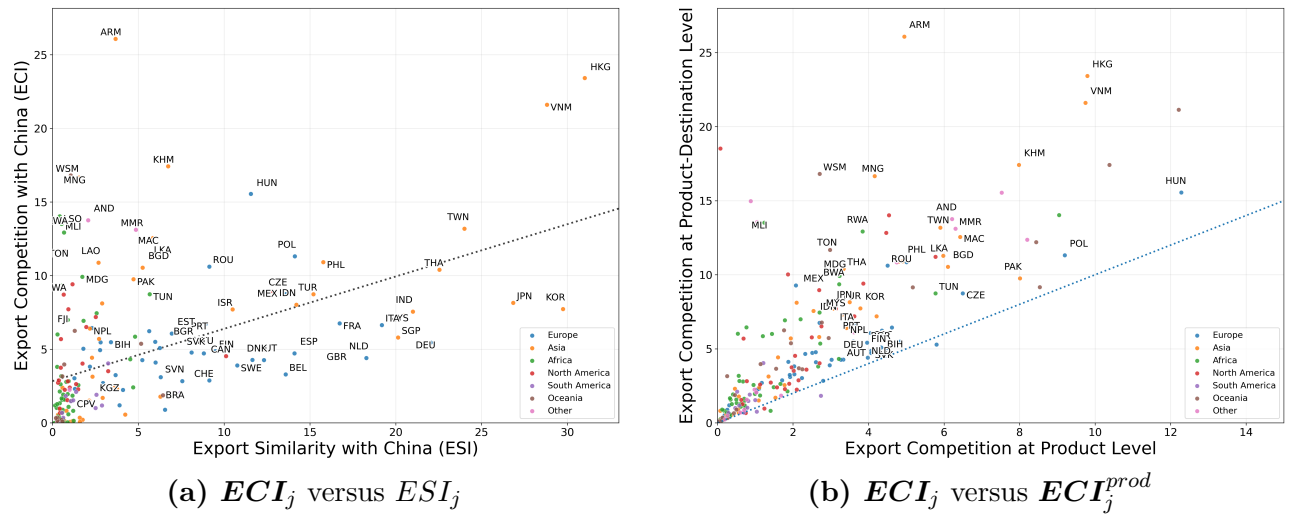
¹⁰For similar reasons, Chinese domestic sales are not included in the denominator of Φ_p in equation (6), although this is not necessarily an issue if one interprets the Chinese export supply elasticity to be determined independently of Chinese domestic supply (see Appendix A.2).

Philippines, and Poland) appear in both lists, but many of the prominent Table 1 countries do not appear among the top-20 in Table 3. Having said this, our index still predicts non-negligible drops in exports for some of the largest economies in the world. For instance, the exports of Mexico, Turkey, Japan and Indonesia are all predicted to drop between 0.80% and 0.90%.

The differences between ESI_j and ECI_j stem from the fact that the two measures capture different objects. ESI_j is an *overlap* index: it is high when a country’s export shares across product–destination cells look similar to China’s. This naturally elevates large, diversified exporters that “look like China” in terms of what they sell and where they sell it. By contrast, ECI_j is an *exposure-to-competition* measure: it loads heavily on where the country actually sells and how intense Chinese competition is in those markets. In particular, a country can have a high ECI_j if it is specialized in products and destinations where China has high baseline penetration in absorption (i.e., China sells a lot in those destination markets), even if the country’s overall export structure is not especially similar to China’s. This mechanism is especially relevant for smaller economies: because their exports are often concentrated in a narrow set of product–destination cells, they can be disproportionately affected by a China shock precisely when those cells coincide with markets in which China is a major supplier. Importantly, this can occur even when those product–destination cells are not large in the aggregate composition of Chinese exports; what matters for competitive pressure is China’s *market share in the destination* and the country’s *own concentration* in that market, not whether the market represents a large fraction of China’s total exports.

Panel (a) of Figure 10 plots our export competition index (ECI) against the export similarity index (ESI). Despite the differences flagged above, the two indices feature a positive correlation of 0.495.

Figure 10: Export Competition Index: Comparisons



In addition to the baseline index in equation (10), we also compute a coarser version of our export competition index at the product level (aggregating the data first across destinations), which is given by

$$\mathbf{ECI}_j^{prod} = 100 \times \sum_{p \in \mathcal{P}} (\tilde{\sigma}_{jp} - 1) \times \frac{\sum_{d \in \mathcal{W}_-} X_{jdp}}{X_j} \times \frac{\sum_{d \in \mathcal{W}_-} X_{cdp}}{\sum_{o \in \mathcal{W}} \sum_{d \in \mathcal{W}_-} X_{odp}} \times \Phi_p, \quad (11)$$

where $\tilde{\sigma}_{jp}$ is the (export-weighted) average demand elasticity for good p faced by country j . This alternative index retains the structure of our preferred, theoretically-grounded index, but it captures better situations in which a large enough shock (e.g., Liberation Day) may lead to large increases in Chinese exports to destinations in which it sold relatively less in the baseline year, something that is inconsistent with the effects of a first-order shock in our model.

Panel (b) of Figure 10 plots this alternative export competition index against our baseline one.¹¹ The rank correlation between the two indices is high (0.862), but there are some notable differences. In particular, Bangladesh and many Eastern European economies (including the Czech Republic, Poland and Hungary) appear much more exposed. These results are in line with those obtained under the export similarity measure (see panel (a) of Figure 9).

On our web interface, we also explore the robustness of our export competition index to the use of a more basic measure of the price shock (ϕ_p instead of Φ_p) and to the ‘long’ variant using market shares computed from 2018 data and tariff differences from January 2018 to September 2025. We label these as $\mathbf{ECI}^{basic}$ and $\mathbf{ECI}^{long}$. The rank correlation between $\mathbf{ECI}$ and $\mathbf{ECI}^{basic}$ is 0.940, while that between $\mathbf{ECI}$ and $\mathbf{ECI}^{long}$ is 0.971.

3.3 Diagnosing Import Exposure

Bystanders of the current U.S.–China trade war may view the constraints faced by Chinese exporters in selling to the United States not only as a competitive threat in export markets, but also as a competitive threat in their *own* market. This reflects a more standard import competition shock, which in this case is triggered by developments in the United States, not China. Of course, the flip side of these import competition forces is the fact that cheaper Chinese imports will reduce the prices faced by consumers, which is a positive aspect of that trade deflection. For this type of import threats or opportunities, neither export similarity between a given country j and China nor our export competition index are particularly informative.

¹¹This alternative index can be downloaded from our website.

Import Similarity

We begin by exploring an index of the similarity between the products that both country j and the U.S. import from China in a baseline year. The idea is that if a given country j imports the same types of goods from China as the U.S., it is more likely that the U.S.–China trade war will result in a surge of Chinese exports to that market j . More specifically, we define this import similarity index (ISI) as

$$ISI_j = 100 \times \sum_{p \in \mathcal{P}} \min \left\{ \frac{M_{jcp}}{M_{jc}}, \frac{M_{ucp}}{M_{uc}} \right\}, \quad (12)$$

where M_{jcp} are country j 's imports of good p from China, M_{ucp} are U.S. imports of good p from China, and M_{jc} and M_{uc} are aggregate imports from China of country j and of the U.S., respectively. We again use 2023 as the reference year for all these flows.

A limitation of the above index is that it may not properly reflect exposure faced by a country that imports certain goods (though not necessarily from China), and these goods happen to be goods that the U.S. was disproportionately importing from China before Liberation Day. The following alternative import similarity index may better capture these situations:

$$ISI_j^{prod} = 100 \times \sum_{p \in \mathcal{P}} \min \left\{ \frac{\sum_{o \in \mathcal{W}} M_{jop}}{M_j}, \frac{M_{ucp}}{M_{uc}} \right\}. \quad (13)$$

The key difference is that product p 's market share for country j is now computed aggregating across origin countries (and not just based on imports from China). The idea is that, if the goal is to predict a surge of imports into country j , it may be better to rely on information on the type of products in which China is more likely to suffer from excess supply, regardless of whether China was already exporting those goods to country j in the baseline year.

A remaining drawback of the above two measures is that they may not capture exposure stemming from Chinese exporters redirecting (or “dumping”) into third-country markets certain products with limited exports to the United States in the benchmark year. This can occur in sectors where Chinese capacity has expanded rapidly but direct exports to the United States have been constrained by U.S. tariffs and other regulatory barriers since the onset of the trade war (if not earlier). Electric vehicles (EVs) illustrate the point: even though Chinese EV sales in the U.S. remain limited, they have sparked significant concern in Europe, despite the consumer gains that would be associated with cheaper EVs. To some extent, these concerns are linked to the U.S.–China trade war: had trade between the two economies remained more liberalized, U.S. demand could have absorbed part of China's surging EV supply, reducing the incentive to push those exports into other markets. To capture these situations, we suggest the following

index:

$$ISI_j^{broad} = 100 \times \sum_{p \in \mathcal{P}} \min \left\{ \frac{M_{jcp}}{M_j}, \frac{X_{cp}}{X_c} \right\}, \quad (14)$$

which captures similarity between the set of products country j imports from China and the set of goods China exports to *all* destinations (not just the U.S.) in the baseline year.¹²

Table 4 lists the 20 economies that, according to our import similarity index in equation (12), may face the highest import exposure because they import from China the type of goods the U.S. imported from China in the baseline year. This list is quite different from that in Table 1 and it is dominated by European countries; it includes only one Asian economy (Japan), while some of the most impacted countries, such as Canada or Finland, were less affected by export competition.

Table 4: Top 20 economies by Import Similarity (ISI_j)

Rank	Country	ISI_j	Rank	Country	ISI_j
1	Canada	70.06	11	Spain	56.77
2	United Kingdom	66.94	12	New Zealand	56.70
3	France	65.09	13	Croatia	55.81
4	Germany	60.42	14	Switzerland	55.66
5	Japan	59.97	15	Estonia	55.66
6	Netherlands	59.83	16	Norway	55.51
7	Finland	59.46	17	Czechia	54.22
8	Poland	58.69	18	Italy	53.06
9	Austria	57.21	19	Latvia	52.72
10	Australia	56.96	20	Sweden	51.97

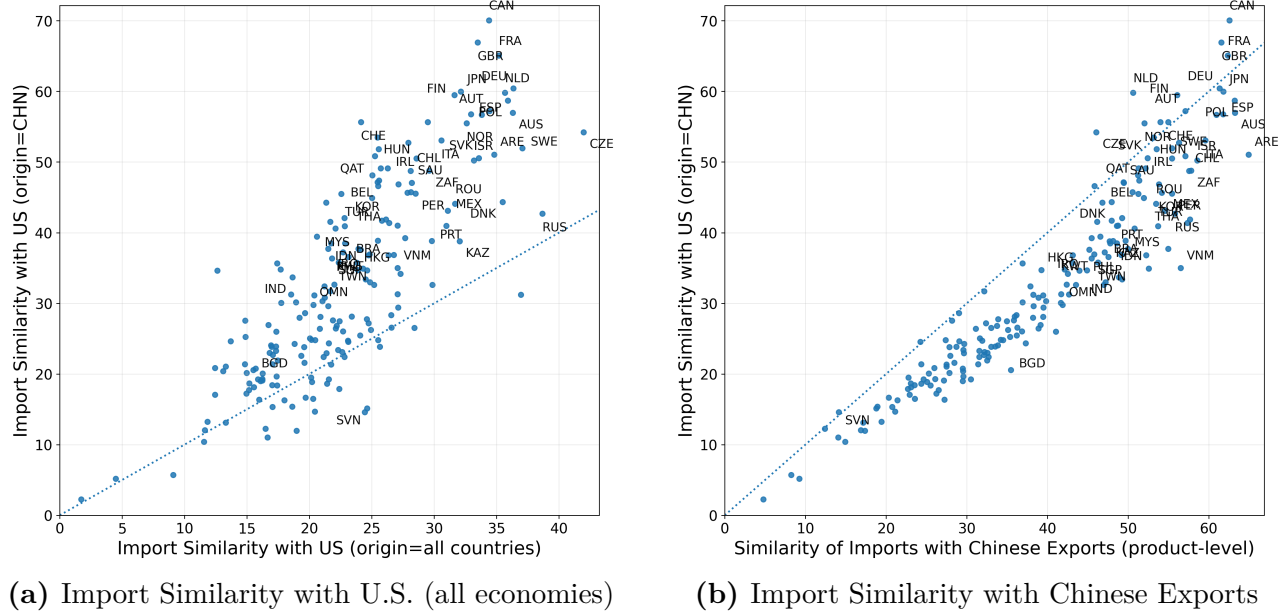
Figure 11 plots our benchmark import similarity measure against the two alternative measures developed in equations (13) and (14). In panel (a), we consider an import similarity index that compares the products China exported to the US in the baseline year with the products imported by country j from all countries (and not just China). The correlation between the two series is high (0.81), but most countries have a higher import similarity vis à vis the U.S. when focusing on imports originating from China, indicating that the possibility that China starts exporting to those countries (or ramps its exports of) products it previously sold to the U.S. does not constitute a notable additional source of exposure for most countries.

When exploring our second alternative import similarity index in equation (14), we find similar results. The correlation between this alternative index and our baseline one is very

¹²Similarly, one could also envision an even more agnostic measure in which the first element of the min operator is replaced with $\sum_{o \in \mathcal{W}} M_{jop}/M_j$ as in equation (13).

high (0.94), so the set of most impacted countries listed in Table 4 is largely unaffected. We do observe, however, that for the vast majority of countries, our baseline import similarity measure might underestimate exposure related to China potentially increasing their exports of certain goods that did not feature prominently in U.S. imports (such as EVs) but are significant for the imports of those economies. The differences, however, are modest.

Figure 11: Alternative Measures of Import Similarity



An Import Competition Index

The above import similarity measures are informative, but they have at least two limitations. First, they are at best measures of ‘exposure’, so it is hard to disentangle whether they capture import competition forces – which reduce the sales and employment of domestic firms – or they reflect potential consumer opportunities derived from access to cheaper imported goods. Second, as with the Finger-Kreinin export similarity index, these indices are not theoretically grounded, so it is not clear that the import shares in equation (12), as well as the min operator in the index, are the relevant objects for assessing import competition threats or cheaper import opportunities. In Appendix A.2, we use our Armington-CES model to shed light on these issues.

We begin by constructing an index of import competition. In particular, we seek a measure of the extent to which the increased Chinese exports diverted into a market j will reduce the size of the domestic, import-competing sector in that country j . Naturally, there is a clear connection between this question and our derivations above related to how Chinese exports were crowding out third-market exporters in export markets. In this case, the destination market

under study coincides with the ‘origin’ country j under consideration, and the relevant reduced sales X_{jjp} correspond to domestic sales of good p by producers in j , which we simply denote by S_{jp} . The resulting import competition index, when expressed relative to aggregate domestic sales (i.e., $\sum_{p \in \mathcal{P}} S_{jp}$) reduces to

$$\mathbf{ICI}_j = \sum_{p \in \mathcal{P}} (\sigma_{op} - 1) \times s_{jp} \times \chi_{cjp} \times \Phi_p, \quad (15)$$

where $s_{jp} = S_{jp} / \sum_{p' \in \mathcal{P}} S_{jp'}$ is the share of domestic sales accounted for by sector p domestic producers.

The intuition for this expression is analogous to that behind our export competition index. The terms Φ_p and χ_{cjp} reflect the relative size of the shock in product p , and how it will be felt in country j , given how much that country relies on Chinese imports of good p in the baseline year. Import growth of product p will be higher when the product $\Phi_p \times \chi_{cjp}$ is larger. The term s_{jp} then captures how this import surge will affect domestic sales, and corresponds to the relative size of sector p in the overall domestic sales in country j . The predicted import growth may be large, but if there are no (or close to no) domestic producers of that good, the resulting decline in domestic sales will be negligible and does not pose a significant threat for the country.

A difficulty in implementing the import competition index in (15) is that it requires information on domestic sales at the product level, which are not readily available to researchers, at least in a unified dataset covering many countries. With that in mind, we compute $\mathbf{ICI}_j$ proxying the domestic share $s_{jp} = S_{jp} / \sum_{p' \in \mathcal{P}} S_{jp'}$ with the share of overall exports of country j that are accounted by exports of good p , or

$$s_{jp} \simeq s_{jp}^X = \frac{\sum_{d \in \mathcal{W}} X_{jdp}}{\sum_{d \in \mathcal{W}} \sum_{p' \in \mathcal{P}} X_{jdp'}}. \quad (16)$$

The idea is that countries will tend to feature higher exports in sectors that are disproportionately large in the industrial structure of a country, and vice versa. Naturally, variation in the tradability of different products introduces noise in our proxy, so our results below should be treated with a grain of salt. Researchers interested in assessing import competition pressures in specific countries should definitely use those product-level domestic sales when that data is available. For similar data limitation issues, and as in the export competition index, we continue to approximate χ_{cop} with the share of China in country o ’s imports of good p .

We compute our import competition index $\mathbf{ICI}_j$ in equation (15) for all countries in our sample for the year 2023. Table 5 ranks countries by this index $\mathbf{ICI}_j$. The top of the distribution is again dominated by smaller and/or more specialized economies (e.g., Bangladesh, Haiti, Vietnam, Cambodia, Myanmar, and Pakistan), followed by a mix of Asian economies and a few small countries. This pattern is consistent with $\mathbf{ICI}$ being an exposure-to-competition

measure: it is high when (i) Chinese import penetration is large in the products affected by the shock and (ii) those products represent a meaningful share of the destination country’s domestic, import-competing sales.

Table 5: Top 20 economies by Import Competition Index (ICI_j)

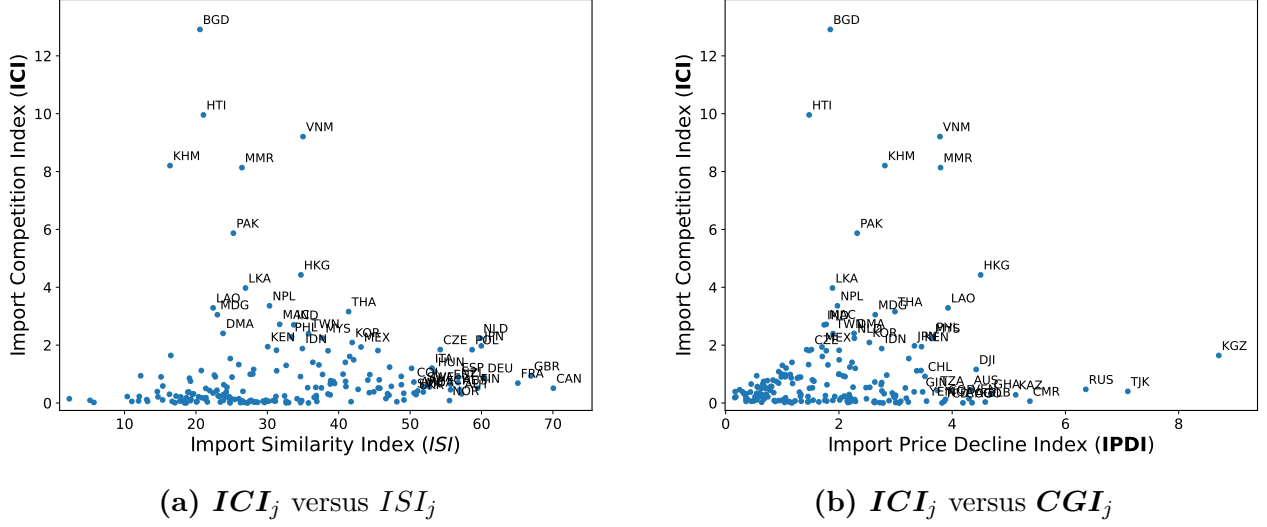
Rank	Country	ICI_j	Rank	Country	ICI_j
1	Sao Tome and Principe	45.16	11	Cambodia	1.69
2	Morocco	16.93	12	Tunisia	1.65
3	Comoros	5.22	13	United States	1.48
4	Vanuatu	4.06	14	Madagascar	1.42
5	Central African Republic	3.11	15	Japan	1.42
6	Vietnam	3.11	16	Dominica	1.41
7	Bangladesh	2.66	17	Pakistan	1.37
8	Guinea	2.04	18	Myanmar	1.36
9	South Africa	2.04	19	Rwanda	1.30
10	Niger	1.80	20	Thailand	1.18

This ranking differs sharply from an import similarity index (ISI), which is an overlap measure based on the composition of a country’s imports (typically relative to the set of goods the U.S. imported from China in the baseline year). ISI therefore tends to rank large, high-income import markets whose import baskets are “U.S.-like” in product composition. By contrast, ICI is designed to capture the *incidence on domestic producers*: it weights product-level import pressure by the importance of each sector in domestic sales. As a result, countries can have high ICI even if they are not particularly similar to the U.S. in what they import from China; what matters is that domestic producers are concentrated in sectors where Chinese import pressure rises. Consistent with the different objects being measured, the overlap between the top-20 lists for ICI and ISI is very limited (in our data, only the Netherlands appears in both top-20 lists), and the raw cross-country correlation between ICI and the baseline ISI measure is close to zero (0.004), as illustrated in panel (a) of Figure 12.

A Consumer Gains Index

As shown above, the Armington-CES model can be used to assess the implications of the U.S.–China trade war for firm sales. Interestingly, it can also be employed to evaluate its consequences for consumer prices. In Appendix A.2, we show that the percentage decline in import prices in a given market j can be approximated, up to a first order, with the following

Figure 12: Import Similarity, Import Competition and Import Price Decline Indices



expression

$$CGI_j = 100 \times \sum_{p \in \mathcal{P}} s_{jp}^M \times \chi_{cjp}^M \times \Phi_p, \quad (17)$$

where Φ_p is identical to that in the expressions for ECI_j and ICI_j in equations (10) and (15), and where χ_{cjp}^M is the Chinese share in total imports of good p in country j (which was already our proxy for χ_{cjp} when empirically implementing our export and import competition indices). The term s_{jp}^M in the above equation for CGI_{jc} is the share of country j 's imports accounted for by sector p . Intuitively, up to scalars, the predicted price decline of imports of good p is proportional to the surge in imports of that good (which is captured by the product $\Phi_p \times \chi_{cjp}^M$), and then the spending shares that are relevant to compute the aggregate consumer gains associated with those lower import prices are the import shares s_{jp}^M .

Table 6 ranks countries by the predicted import-price decline from a China-specific supply shock, CGI_j . To reiterate, this measures mechanically puts weight on (i) how large China is in a country's import basket in the affected products (χ_{cjp}^M) and (ii) how important those products are in the country's overall import spending (s_{jp}^M).

Compared to Table 5, which captures producer-side exposure and therefore loads on the importance of the affected sectors in domestic, import-competing activity rather than in import expenditures), the list in Table 6 tilts toward economies that are highly import-dependent on China in the relevant sectors, so it is not surprising to see several smaller, import-reliant economies (and a few larger ones with sizable China import shares in the impacted goods, such as Australia) rise to the top, even if their domestic producers are not especially concentrated in those sectors. By contrast, Table 5 features economies where the shock maps strongly into domestic producer displacement because the affected sectors represent a meaningful share of

Table 6: Top 20 economies by Consumer Gains Index (CGI_j)

Rank	Country	CGI_j	Rank	Country	CGI_j
1	Tajikistan	1.57	11	Congo - Brazzaville	1.03
2	Tonga	1.32	12	Benin	1.01
3	Cameroon	1.32	13	Jamaica	1.01
4	Venezuela	1.29	14	Angola	1.01
5	Djibouti	1.23	15	Belarus	1.00
6	Russia	1.21	16	Congo - Kinshasa	0.98
7	Chad	1.20	17	Somalia	0.97
8	Solomon Islands	1.09	18	Equatorial Guinea	0.97
9	Comoros	1.08	19	Sudan	0.96
10	Eritrea	1.04	20	Yemen	0.95

domestic sales. The limited overlap between the two tables (e.g., Hong Kong SAR and a few Asian economies) is thus exactly what the theory implies: consumer price gains are governed by import-expenditure weights, whereas producer pressures are governed by the domestic sectoral footprint of the same goods.[**EDIT WHEN FINAL TABLES ARE AVAILABLE**]

In panel (b) of Figure 12 we plot ICI_j and CGI_j against each other for all countries in our sample. The correlation between the two series is positive but very small (0.133).[**EDIT WHEN FINAL TABLES ARE AVAILABLE**]

Table 7: Top 20 economies by Consumer Gains Index (Downstream)($CGI_j^{downstream}$)

Rank	Country	$CGI_j^{downstream}$	Rank	Country	$CGI_j^{downstream}$
1	Djibouti	1.92	11	Angola	1.41
2	Tajikistan	1.82	12	Japan	1.38
3	Cameroon	1.78	13	Solomon Islands	1.35
4	Togo	1.76	14	Jamaica	1.34
5	Liberia	1.58	15	Tonga	1.34
6	Comoros	1.48	16	Yemen	1.34
7	Venezuela	1.46	17	Russia	1.33
8	Vietnam	1.43	18	Congo - Brazzaville	1.33
9	Chad	1.42	19	Benin	1.28
10	Puerto Rico	1.41	20	Marshall Islands	1.27

[**ADD CGI FOR FINAL GOODS**]. Operationally, we distinguish inputs from final goods using the value-chain position of products in the spirit of Antràs et al. (2012): products

that are more *upstream* (i.e., typically used as inputs into further production stages before reaching final demand) are classified as intermediates, whereas more *downstream* products are closer to final absorption. We label product p as a final good if its upstreamness measure is lower than 2. We define the set of final products by $\mathcal{P}^F \subseteq \mathcal{P}$.¹³

3.4 Diagnosing Sourcing Exposure

A distinctive feature of the current U.S.–China trade war is that the excess Chinese supply displaced from the U.S. market need not be concentrated in final or consumption goods. A substantial share of the affected products are *intermediate inputs* – parts, components, and materials used by firms in subsequent stages of production (cf., [Bown and Zhang, 2019](#)). When Chinese exports of such inputs are redirected toward third markets, the resulting shock has a fundamentally different interpretation than in the case of final goods: it may intensify import competition for domestic input producers, but it can also relax sourcing constraints and reduce costs for domestic downstream users of those inputs, thereby strengthening their competitiveness.

It is straightforward to develop a version of our import competition index in equation (15) that restricts attention to intermediate inputs. We do so in Appendix A.2 and calculate it for all countries in Appendix XYZ. In the main text, we focus on developing an index of the sourcing gains associated with cheaper Chinese intermediate inputs. This is not simply a version of our consumer gains index restricted to intermediate inputs because properly accounting for sourcing gains requires the use of information on input-output links between sectors.

A Sourcing Gains Index

To account for the gains from cheaper inputs, we develop a variant of our Armington-CES model that incorporates input-output links. In particular, production of each good p now uses inputs of product-type z with a cost share a_{pz} . Cheaper intermediate inputs generate heterogeneous effects on downstream sectors depending on the drop in prices of the particular inputs that each downstream sector uses. To aggregate these effects, we trace the implications of these lower costs for the exports of a country’s exports to all destinations (other than the U.S. and China). In Appendix A.2, we then show that the percentage response of country o ’s exports to Chinese cheaper intermediate inputs is given by

$$SGI_j = 100 \times \sum_{d \in \mathcal{W}_-} \frac{X_{jd}}{X_j} \sum_{p \in \mathcal{P}} (\sigma_{dp} - 1) \times s_{jdp} \times \sum_{z \in \mathcal{P}^I} a_{pz} \times \chi_{c_j z}^M \times \Phi_z. \quad (18)$$

¹³In the implementation, this requires mapping the upstreamness measure to our HS product classification and defining the input set $\mathcal{P}^F \subseteq \mathcal{P}$ at the HS6 level.

[DONE UP TO HERE]

$$\mathbf{RPI}_{ciu} \equiv \sum_{p \in \mathcal{P}} s_{iup} \left(\sum_{z \in \mathcal{P}^I} a_{pz} \times \Phi_z \times \chi_{ciz}^M \right). \quad (19)$$

show that the percentage fall in imported input prices in a given destination market d can be approximated, up to a first order, with the following expression

$$\mathbf{SPDI}_{dc} = 100 \times \sum_{p \in \mathcal{P}^I} \Phi_p \times \chi_{cdp}^M \times s_{dp}^{M,I}, \quad (20)$$

where Φ_p and χ_{cdp}^M are (again) identical to those employed in the previous indices, and where $s_{dp}^{M,I}$ is the share of country d 's imports of intermediate inputs accounted for by sector p . In the model in Appendix A.2, the term $\mathbf{SPDI}_{dc}$ captures (up to a scalar common across countries) the percentage decline in intermediate input import prices associated with the U.S. policy that has rendered Chinese exports of inputs cheaper in other markets.

Table 8 reports the Sourcing Price Decline Index ($\mathbf{SPDI}_{dc}$), led by Tonga, the Kyrgyz Republic, Russia, Tajikistan, and the Solomon Islands (with Zimbabwe, Cuba, and Montenegro also near the top). Compared with the sourcing competition index $\mathbf{SCI}$ in Table ?? – whose top countries are Bangladesh, Vietnam, Myanmar, Cambodia, and Tonga – $\mathbf{SPDI}$ shifts attention from *producer competition* to *import-cost exposure*. Economically, $\mathbf{SPDI}$ is high where a country's imported intermediate-input basket is concentrated in products for which China's predicted price declines (and/or China's supply presence) are particularly large, so the main effect is cheaper inputs for downstream firms rather than displacement of domestic intermediate producers. That is why $\mathbf{SPDI}$ highlights several small, import-dependent economies and a few countries with concentrated intermediate import profiles, while $\mathbf{SCI}$ highlights places where domestic upstream producers are most likely to be challenged. The overlap for Tonga (top in $\mathbf{SPDI}$ and also top-5 in $\mathbf{SCI}$) and Cambodia (top-10 in both) is consistent with some economies facing both forces simultaneously: tougher upstream competition alongside cheaper imported inputs for downstream production. Panel (b) of Figure ?? plots $\mathbf{SCI}_{dc}$ against $\mathbf{SPDI}_{dc}$ and shows that there is virtually no correlation between the two series (-0.005).

3.5 Diagnosing Rerouting

Our final *rerouting* channel captures a distinct margin of adjustment to U.S.–China trade frictions: production (or, more mechanically, recorded trade flows) shifts away from China and toward third countries that become platforms for serving the U.S. market. In its “real” form, this reflects relocation of downstream stages – often final assembly – to a third country, while upstream stages and component production remain in China. In its “statistical” form, it

Table 8: Top 20 economies by $\mathbf{SGI}_{dc}$

Rank	Country	$\mathbf{SGI}_j$	Rank	Country	$\mathbf{SGI}_j$
1	Mozambique	6.47	11	Libya	1.73
2	Djibouti	4.15	12	Kiribati	1.67
3	Chad	2.87	13	South Africa	1.52
4	Vanuatu	2.57	14	Kazakhstan	1.50
5	Nigeria	2.55	15	Zambia	1.43
6	Micronesia	2.53	16	Russia	1.42
7	Sudan	2.47	17	Palau	1.42
8	Kuwait	2.16	18	Azerbaijan	1.36
9	Tonga	2.03	19	Brazil	1.31
10	Brunei	1.84	20	Malaysia	1.27

reflects re-exporting/transshipment or limited transformation, whereby goods enter the U.S. after passing through an intermediary location.

This mechanism does not simply reflect *sourcing opportunities*, according to which cheaper Chinese intermediates can improve competitiveness broadly, potentially raising exports across *all* destinations. Instead, under rerouting, the salient prediction is *destination-specific*: countries that expand exports to the U.S. in sectors where they are simultaneously deepening linkages to Chinese supply chains (whether through inputs or even the same HS product) are those most exposed to rerouting dynamics.

Import-Export Similarity

To operationalize this idea, we first construct an Import–Export Similarity Index (IESI) that compares (i) the composition of country i ’s exports to the U.S. with (ii) the composition of country i ’s imports from China. When computed at a fine product level (e.g., HS-6), the index is informative about rerouting of very similar products (including potential re-exporting). When computed at a more aggregated level (e.g., say HS-4 or HS-2 levels), it is better suited to capture relocation patterns in which China continues to supply parts and components while final-stage production is performed elsewhere.¹⁴ Formally, we define:

$$IESI_{ciu} = 100 \times \sum_{p \in \mathcal{P}} \min \left\{ \frac{X_{iup}}{X_{iu}}, \frac{M_{icp}}{M_{ic}} \right\}, \quad (21)$$

¹⁴When available, direct information on input-output links would naturally lead to more precise indices of sourcing and rerouting opportunities.

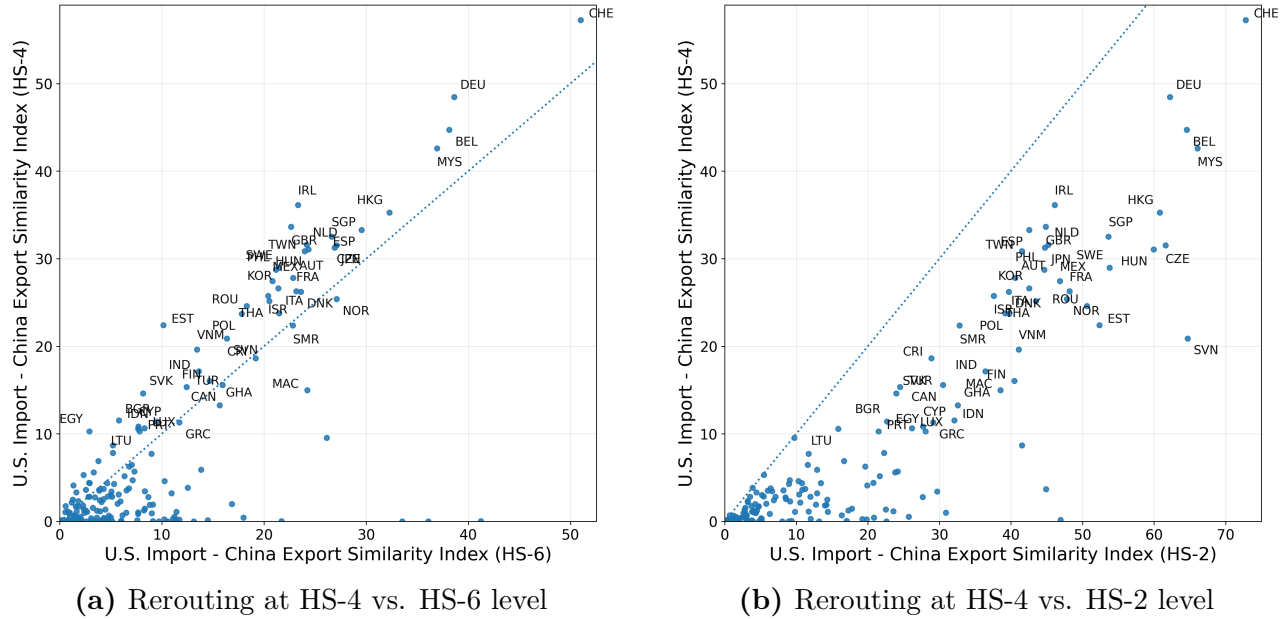
where X_{iu} are country i 's total exports to the U.S. and M_{ic} are country i 's total imports from China. As a baseline, we compute this Import–Export similarity index at the HS-4 level (with 2023 data), but we will explore its robustness to alternative disaggregation levels.

Table 9 lists the 20 economies with the highest amount of rerouting according to our similarity index in equation (21). The list includes a few economies – Hong Kong SAR, Taiwan Province of China, or Mexico – that have been mentioned in recent years as “connectors” between China and the U.S., but it also includes a number of Asian and European countries that, according to our index, have the potential to serve as “intermediaries” of this bilateral relationship in future years.

Table 9: Top 20 economies by Import-Export China-U.S. Similarity ($IESI_{ciu}$)

Figure 13 plots our benchmark HS-4 similarity index against two alternative indices computed at a higher disaggregation level (HS-6 in panel (a)) and at a lower disaggregation level (HS-2 in panel (b)). The results are qualitatively similar in both cases, with correlations of 0.89 and 0.80 in panel (a) and (b), respectively. As a result, the top countries in terms of rerouting potential are the same ones according to all three measures.

Figure 13: Rerouting at Alternative Disaggregation Levels



Rerouting Potential Index

In addition to the Import–Export Similarity Index, we construct a theoretically-motivated rerouting index that links cheaper Chinese intermediate inputs to the potential expansion of country i ’s exports to the U.S. in the same sectors. The key idea is that rerouting is a *destination-specific* adjustment margin: countries are more likely to expand exports to the U.S. in sectors in which their marginal costs fall because they source intermediate inputs from China.

In principle, the marginal-cost reduction for an exported product p depends on the full input–output (IO) structure. In Appendix A.2, we show that, under Armington CES demand in the U.S. and a China-specific price shock, the percentage change in exports of product p from i to the U.S. is proportional (up to common elasticities and pass-through parameters) to the fall in the intermediate input price index relevant for producing p in i . This yields the IO-based rerouting potential index

$$\mathbf{RPI}_{icu} \equiv 100 \times \sum_{p \in \mathcal{P}} s_{iup} \left(\sum_{z \in \mathcal{P}^I} a_{pz} \times \Phi_z \times \chi_{ciz}^M \right), \quad (22)$$

where $s_{iup} \equiv X_{iup}/X_{iu}$ is the baseline share of product p in country i ’s exports to the U.S., χ_{ciz}^M is China’s share in country i ’s imports of intermediate input z , Φ_z is the China-only shock term defined above, and a_{pz} are IO weights mapping output p into its intermediate input requirements.

In the absence of a harmonized IO table at the HS level, we propose a set of trade-based approximations that impose structure on a_{pz} using HS nesting. For $\ell \in \{6, 4, 2\}$, define the HS- ℓ neighborhood of output good p as $\mathcal{P}^{I,(\ell)}(p) \equiv \{z \in \mathcal{P}^I : HS_\ell(z) = HS_\ell(p)\}$, and proxy a_{pz} by country i ’s baseline intermediate import shares within that neighborhood:

$$s_{iz|p}^{M,I,(\ell)} \equiv \frac{M_{iz}^I}{\sum_{z' \in \mathcal{P}^{I,(\ell)}(p)} M_{iz'}^I}, \quad q \in \mathcal{P}^{I,(\ell)}(p).$$

This yields a product-specific sourcing decline index,

$$\mathbf{SSDI}_{icp}^{(\ell)} \equiv 100 \times \sum_{q \in \mathcal{P}^{I,(\ell)}(p)} \Phi_q \times \chi_{ciq}^M \times s_{iq|p}^{M,I,(\ell)},$$

and an empirically implementable rerouting potential index,

$$\mathbf{RPI}_{icu}^{(\ell)} \equiv \sum_{p \in \mathcal{P}} s_{iup} \times \mathbf{SSDI}_{icp}^{(\ell)}. \quad (23)$$

The three levels of ℓ correspond to progressively broader assumptions about input linkages:

HS-6 captures “own-product” rerouting (including re-exporting), while HS-4 and HS-2 allow for parts-and-components linkages within a narrower or broader industry. Finally, if all outputs were to use a common economy-wide input bundle, then $\mathbf{SSDI}_{icp}^{(\ell)}$ would collapse to an aggregate sourcing price decline measure and the export composition weights s_{iup} would no longer matter for the predicted percentage change in X_{iu} .¹⁵

Table 10 reports the top-20 countries in terms of our Rerouting Potential Index ($\mathbf{RPI}$). The ranking is led by Laos, Nepal, Haiti and Bangladesh (with Cambodia and Vietnam also near the top). The limited overlap between this list and that for the Import-Export Similarity Index in Table 9 shows is intuitive: IESI is a *pure composition-overlap* statistic between what a country exports to the U.S. and what it imports from China, whereas RPI further scales that overlap by where the model predicts the China-to-U.S. sourcing reallocation/price pressure is most intense across HS4 sectors. As a result, countries with highly concentrated export baskets and strong reliance on Chinese inputs in a few “high-rerouting” sectors can rise sharply in the $\mathbf{RPI}$ ranking even if their overall overlap is not among the very highest in $IESI$. Where the lists *do* intersect (e.g., Hong Kong SAR, Taiwan Province of China), it flags economies that are simultaneously tightly linked to Chinese inputs and have U.S.-oriented exports in the same industries, making them natural candidates for trade rerouting effects.

Table 10: Top 20 economies by $\mathbf{RPI}_{ciu}$ (HS4)

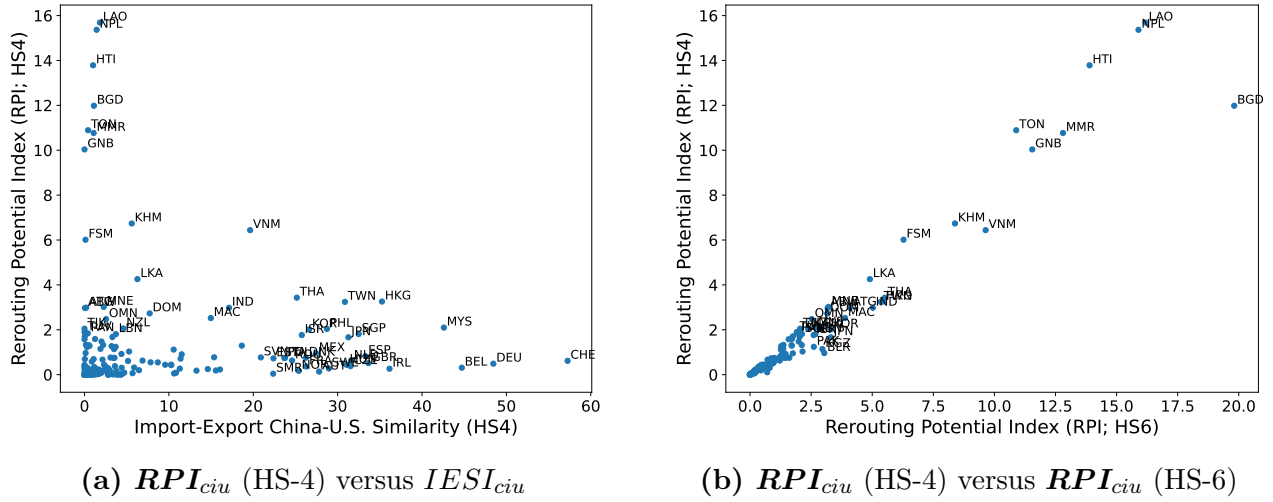
Panel (a) and (b) of Figure 14 plot our baseline $\mathbf{RPI}_{ciu}$ index computed at the HS-4 level against the import-export similarity index $IESI_{ciu}$ and the $\mathbf{RPI}_{ciu}$ index computed at the HS-6 level, respectively. Panel (a) shows close to no correlation (-0.040) between $\mathbf{RPI}_{ciu}$ and $IESI_{ciu}$, while the level of disaggregation of the $\mathbf{RPI}_{ciu}$ seems less material, as the correlation between the two series in panel (b) is very high (0.949).

4 Conclusions

The post-2018 period is best understood not as a generalized reversal of globalization, but as a reorganization of trade and production linkages in response to policy shocks and strategic rivalry. Using up-to-date information on the evolving tariff regime and recent trade flows, we have documented that the U.S.–China trade war has generated a sharp reallocation of U.S. import sourcing away from China and a corresponding reorientation of Chinese exports

¹⁵We have also computed a version of the $\mathbf{RPI}$ index that includes *all* goods and not just intermediate inputs in constructing the price decline component of the index. Although there are some differences in the top countries under both measures, the correlation between the two indices is high (0.56).

Figure 14: Import-Export Similarity and Rerouting Potential



toward other destinations, with an acceleration in adjustment during the most recent escalation following Liberation Day. These shifts have reshaped competitive conditions globally, even when aggregate world trade remains broadly stable.

A central message of the paper is that third-country outcomes cannot be read solely through the lens of simple trade diversion. To discipline an interpretation of these third-market effects, we have proposed a taxonomy that distinguishes six (non-mutually exclusive) margins – export competition, import competition, import price declines, sourcing competition, sourcing price declines, and rerouting potential – and operationalized them with diagnostic indices. This approach clarifies why some economies appear to benefit on one margin while facing intensified pressure on another.

We also highlighted that the current episode is unfolding in an “age of geoeconomics,” where trade policy tools are increasingly used not just to pursue economic objectives but to generate leverage and manage geopolitical risk. Empirically, geopolitical alignment has become more predictive of bilateral trade outcomes, consistent with the rise of “friend-shoring”; at the same time, the most recent U.S. tariff actions appear broad-based, suggesting a more unilateral and transactional deployment of tariffs as a general instrument of leverage. We then brought these developments to the question of whether cooperative trade outcomes can be sustained when policy incentives are shaped by both market power and geopolitical influence.

The analytical framework in Section ?? makes precise a mechanism linking geopolitical influence to the stability of low-tariff cooperation. When influential states can secure relatively favorable outcomes outside agreements – because other governments partially internalize their welfare – those states have less to gain from binding commitments and stronger incentives to act unilaterally. In that environment, the durability of a cooperative trading system becomes

especially sensitive to how geopolitical power is distributed and how tightly it aligns with economic size. Consistent with this mechanism, the evidence reviewed in Section ?? indicates that the empirical alignment between economic size and geoeconomic power has strengthened over time, reinforcing the paper’s broader interpretation of why strains on rules-based cooperation have become more acute.

Looking ahead, our results suggest that restoring a stable, rules-based trading order will require more than marginal tariff rollbacks. Even when world aggregate remains broadly stable, we are likely entering a phase of *fragmented interdependence*, in which the structure of integration is likely to become more segmented, with greater weight on geopolitically aligned partners and on “connector” economies that facilitate reallocation and rerouting.

References

- Aiyar, S., A. F. Presbitero, and M. Ruta (Eds.) (2023, September). *Goeconomic Fragmentation: The Economic Risks from a Fractured World Economy*. CEPR Press. CEPR-IMF eBook (conference proceedings, May 2023).
- Alfaro, L. and D. Chor (2023, August). Global supply chains: The looming “great reallocation”. Technical report, Federal Reserve Bank of Kansas City. Paper prepared for the Jackson Hole Symposium, 24–26 Aug 2023. Draft: 30 August 2023.
- Alfaro, L. and D. Chor (2025, November). An anatomy of the great reallocation in US supply chain trade. NBER Working Paper 34490, National Bureau of Economic Research.
- Almunia, M., P. Antràs, D. Lopez-Rodriguez, and E. Morales (2021, November). Venting Out: Exports during a Domestic Slump. *American Economic Review* 111(11), 3611–62.
- Amiti, M., S. J. Redding, and D. E. Weinstein (2019). The impact of the 2018 tariffs on prices and welfare. *Journal of Economic Perspectives* 33(4), 187–210.
- Antràs, P. (2021). De-globalisation? Global value chains in the post-COVID-19 age. In *ECB Forum on Central Banking, November 2020: Conference proceedings*, pp. 28–80. Frankfurt am Main: European Central Bank. Event held online, 11–12 November 2020.
- Antràs, P., D. Chor, T. Fally, and R. Hillberry (2012). Measuring the upstreamness of production and trade flows. *American Economic Review* 102(3), 412–416.
- Antràs, P. and V. Tubdenov (2025). Measuring the Average Period of Production. In *AEA Papers and Proceedings*, Volume 115, pp. 624–630. American Economic Association 2014 Broadway, Suite 305, Nashville, TN 37203.
- Aprigliano, V., L. Bencivelli, A. Borin, F. P. Conteduca, F. Corsello, S. Federico, C. Giordano, F. Leone, M. Mancini, L. Panon, L. Patrignani, and S. Pica (2026, Jun). China Shock 2.0: structural drivers and implications for the euro area. *Questioni di Economia e Finanza (Occasional Papers)* 1025, Bank of Italy, Economic Research and International Relations Area.
- Bailey, M. A., A. Strezhnev, and E. Voeten (2017). Estimating Dynamic State Preferences from United Nations Voting Data. *Journal of Conflict Resolution* 61(2), 430–456.
- Bown, C. and Y. Zhang (2019). Measuring trump’s 2018 trade protection: Five takeaways. *Peterson Institute Trade and Investment Policy Watch*, 3–108.

- Bown, C. P. (2025, November). US–China trade war tariffs: An up-to-date chart. Peterson Institute for International Economics (PIIE), Chart.
- Cavallo, A., G. Gopinath, B. Neiman, and J. Tang (2021). Tariff pass-through at the border and at the store: Evidence from US trade policy. *American Economic Review: Insights* 3(1), 19–34.
- Chatterjee, S. and A. Subramanian (2026, 5). China’s Mercantilist Squeeze on Developing Countries. Working Paper 26-7, Peterson Institute for International Economics.
- Chowdhry, S., K. Erhardt, and J. Hinz (2026). Competing with China in third markets. Kiel Policy Brief 213, Kiel Institute for the World Economy.
- de Soyres, F., E. Fisgin, A. M. Santacreu, E. V. Leemput, and K. Vega (2026). China shock 2.0: How China’s ongoing export surge differs from the early 2000s. FEDS Notes May 29, Board of Governors of the Federal Reserve System (U.S.).
- Emlinger, C., I. Mejean, V. Vicard, and K. Lefebvre (2026). EU under pressure? Exploring Chinese trade deflection. In R. Helene, B. Weder di Mauro, and J. Zettelmeyer (Eds.), *Paris Report 4: The New Global Imbalances* (1 ed.), Volume 1, Chapter 14. Centre for Economic Policy Research.
- Fajgelbaum, P. D., P. K. Goldberg, P. J. Kennedy, and A. K. Khandelwal (2020). The return to protectionism. *The Quarterly Journal of Economics* 135(1), 1–55.
- Fetzer, T., P. J. Lambert, B. Feld, and P. Garg (2024). Ai-generated production networks : Measurement and applications to global trade. The warwick economics research paper series (twerps).
- Finger, J. M. and M. E. Kreinin (1979). A measure of ‘export similarity’ and its possible uses. *The Economic Journal* 89(356), 905–912.
- Flaaen, A., A. Hortaçsu, and F. Tintelnot (2020). The production relocation and price effects of US trade policy: The case of washing machines. *American Economic Review* 110(7), 2103–2127.
- Gaulier, G. and S. Zignago (2010). BACI: International Trade Database at the Product-Level. The 1994-2007 Version. Working Papers 2010-23, CEPII.
- Goldberg, P. K. and T. Reed (2023). Is the global economy deglobalizing? if so, why? and what is next? *Brookings Papers on Economic Activity* 2023(1), 347–423.

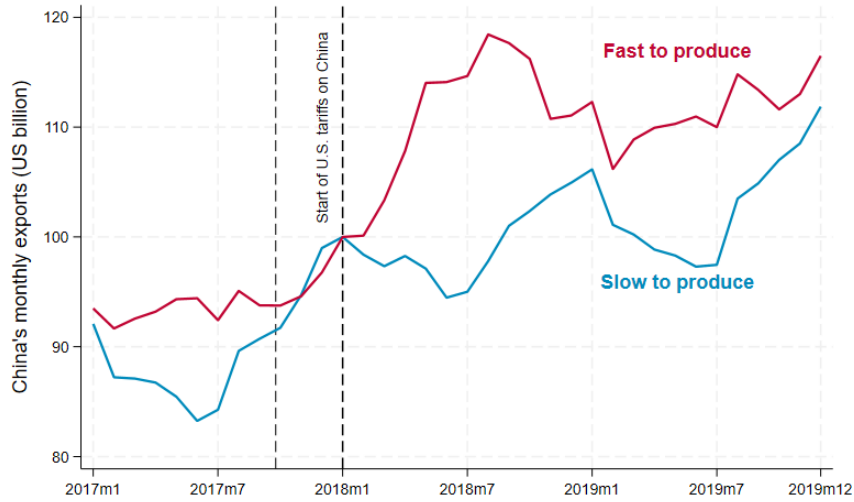
- Gopinath, G., P.-O. Gourinchas, A. F. Presbitero, and P. Topalova (2025). Changing global linkages: A new cold war? *Journal of International Economics* 153, 104042.
- Jenkins, R. (2008). Measuring the competitive threat from china for other southern exporters. *World Economy* 31(10), 1351–1366.
- Liao, S., I. S. Kim, S. Miyano, and H. Zhang (2020). *concordance: Product Concordance*. R package version 2.0.0.
- Malesky, E. J., J. Y. Wen, S.-J. Wu, B. Feng, and E. Iyoha (2024, May). Exports in disguise: Trade re-routing during the U.S.–China trade war. Working Paper 24-072, Center for International Development, Harvard University.
- Rodríguez-Clare, A., F. Teti, M. Ulate, J. P. Vasquez, and R. D. Zarate (2025). The 2025 trade war: Dynamic impacts across us states and the global economy.
- Schulte, P., A. Enders, A. Esser, and F. Strobel (2026). From tariffs to trade flows: Diversion effects and China’s exports to the EU. VoxEU Column, 26 February.
- Schulze, T. and W. Xin (2025, Jun). Demystifying trade patterns in a fragmenting world. IMF Working Papers 2025/129, International Monetary Fund.
- Soderbery, A. (2018). Trade elasticities, heterogeneity, and optimal tariffs. *Journal of International Economics* 114, 44–62.
- Sun, G.-Z. and Y.-K. Ng (2000). The measurement of structural differences between economies: An axiomatic characterization. *Economic Theory* 16(2), 313–321.
- Teti, F. (2024). Missing tariffs.

A Appendix

A.1 The Average Period of Production and Trump 1.0 tariffs

Figure A.1 provides evidence for a novel channel contributing to the gradual nature of the U.S.–China decoupling. More specifically, building on a measure of the duration of production developed by [Antràs and Tubdenov \(2025\)](#) – the so-called average period of production or $\mathcal{APP}$ – Figure A.1 shows that reallocation of Chinese exports away from the U.S. was faster in products with a shorter $\mathcal{APP}$, while products with longer production cycles adjusted more slowly. Indeed, the figure shows that the initial jump in Chinese exports to countries other than the U.S. after the introduction of the first U.S. tariffs in January 2018 was entirely driven by “fast to produce” goods.

Figure A.1: The Great Reallocation and the Temporal Dimension of Production



Notes: Monthly exports are computed as backward-looking 6-month moving averages. We define goods as ‘fast to produce’ if their $\mathcal{APP}$ is in the top quartile of the distribution and we label the others as “slow to produce”.

Sources: Trade Data Monitor, [Bailey et al. \(2017\)](#), [Antràs and Tubdenov \(2025\)](#).

A.2 Microfoundation for our Indices: An Armington-CES Model

Environment The world consists of a finite set of countries indexed by $o, d \in \mathcal{W}$ and a finite set of products indexed by $p \in \mathcal{P}$. We interpret d as a destination (importing) market, while o denotes an origin (exporting) country. Countries may act as both importers and exporters. At times we will refer explicitly to the U.S. (indexed by u) and China (indexed by c).

For each destination d and product p , total nominal expenditure on product p in market d is denoted E_{dp} . Throughout, E_{dp} is treated as a constant in our comparative-static exercises below.

Each origin o supplies a single Armington variety of each product p to each destination d at a delivered price p_{odp} . These delivered prices can be interpreted as incorporating production costs, trade costs, and markups.

Preferences and Demand Fix a destination market d and product p . Consumers in d have CES preferences over country-of-origin varieties of product p :

$$Q_{dp} = \left(\sum_{i \in \mathcal{W}} q_{odp}^{\frac{\sigma_p - 1}{\sigma_p}} \right)^{\frac{\sigma_p}{\sigma_p - 1}}, \quad \sigma_p > 1, \quad (\text{A.1})$$

where q_{odp} is quantity of the i -variety of product p consumed in market m , and σ_p is the elasticity of substitution across origins within product p . The CES price index for product p in destination s is

$$P_{dp} = \left(\sum_{i \in \mathcal{W}} p_{odp}^{1 - \sigma_p} \right)^{\frac{1}{1 - \sigma_p}},$$

where remember that p_{odp} is the delivered price of variety o of good p sold in market d .

Let X_{opd} denote nominal expenditure (revenue) in destination d on product p sourced from origin o :

$$X_{opd} \equiv p_{odp} q_{odp}.$$

CES demand implies that the expenditure share of origin o in destination d and product p is

$$\chi_{opd} \equiv \frac{X_{opd}}{E_{dp}} = \frac{p_{odp}^{1 - \sigma_p}}{\sum_{o' \in \mathcal{W}} p_{o'pd}^{1 - \sigma_p}}, \quad \sum_{i \in \mathcal{W}} \chi_{opd} = 1. \quad (\text{A.2})$$

Thus, bilateral revenues satisfy $X_{opd} = \chi_{opd} E_{dp}$.

A Price Shock and Third-Country Export Losses We are interested in how an exporter-specific price change in a ‘destination–product’ market reallocates expenditure across origins, and thereby affects third-country export revenues.

Fix a particular origin c (e.g., China) and consider a small proportional decline in its delivered price in a given destination d and product p :

$$d \ln p_{cdp} = -\Phi_p, \quad \Phi_p > 0, \quad (\text{A.3})$$

holding fixed (i) total expenditure E_{dp} and (ii) the delivered prices of all other origins, p_{opm}

for $o \neq c$. The shock may be interpreted as a cost decrease, a relative wage decline, or an outward supply shift for exporter c in market (p, d) . We shall return to the latter interpretation in later subsection. Because we only consider this bilateral shock, from the point of view of third markets, we can safely refer to ε_{cdp} as product-level shock, which we denote by Φ_p .

Let o denote a third-country exporter potentially displaced by c in market (p, d) . Since $X_{odp} = E_{dp}\chi_{odp}$, changes in X_{odp} arise through changes in χ_{odp} induced by (A.3). Differentiating the CES share formula yields the standard identity:

$$dX_{odp} = -(\sigma_p - 1) \times \Phi_p \times E_{dp} \times \chi_{odp} \times \chi_{cpd}. \quad (\text{A.4})$$

Equation (A.4) captures the crowding out effect of the increased Chinese exports, i.e., the revenue loss of exporter o in a given market. This loss is proportional to (i) the substitution elasticity $\sigma_p - 1$, (ii) the size of the shock Φ_p (ii) the size of the market E_{dp} , and (iii) the product of the two competitors' initial expenditure shares, $\chi_{odp} \times \chi_{cpd}$. Holding fixed the elasticity and market size, third-country losses are largest in destination-product markets where (a) exporter o is important (high χ_{odp}) and (b) exporter c is also important (high χ_{cpd}). If either exporter has negligible market share, there is little (first-order) displacement in that market.

Aggregation Across Products and Destinations Define total exports of country o as the sum of revenues across all destinations and products:

$$X_o \equiv \sum_{d \in \mathcal{W}} \sum_{p \in \mathcal{P}} X_{odp}.$$

Summing (A.4) across (p, d) yields the first-order change in total exports:

$$dX_o = - \sum_{d \in \mathcal{W}} \sum_{p \in \mathcal{P}} (\sigma_p - 1) \times \Phi_p \times E_{dp} \times \chi_{cpd} \times \chi_{odp}. \quad (\text{A.5})$$

To express (A.5) in terms of observable trade shares, define:

$$X_{od} \equiv \sum_{p \in \mathcal{P}} X_{odp} \quad (\text{exports of } o \text{ to destination } d), \quad (\text{A.6})$$

$$s_{od} \equiv \frac{X_{od}}{X_o} \quad (\text{destination share in } o\text{'s total exports}), \quad (\text{A.7})$$

$$s_{odp} \equiv \frac{X_{odp}}{X_{od}} \quad (\text{product share within } o\text{'s exports to } d). \quad (\text{A.8})$$

Note that $X_{odp} = X_o \cdot s_{od} \cdot s_{odp}$, and that $\chi_{cdp} = X_{cdp}/E_{dp}$ is the share of destination d 's expenditure on product p sourced from China.

The Export Competition Index Dividing the change in country o 's exports by the initial level of exports, we obtain

$$\frac{dX_o}{X_o} = - \sum_{d \in \mathcal{W}} \frac{X_{od}}{X_o} \sum_{p \in \mathcal{P}} (\sigma_p - 1) \times \Phi_p \times \chi_{cpd} \times s_{odp}.$$

If the elasticity of demand does not vary across products, then notice that

$$\mathbf{ECI}_{oc} = \sum_{d \in \mathcal{W}} \frac{X_{od}}{X_o} \sum_{p \in \mathcal{P}} \Phi_p \times \chi_{cpd} \times s_{odp}, \quad (\text{A.9})$$

which corresponds to equation (10) in the main text, except for the set $\mathcal{W}$ in the first summation. A key aspect of $\mathbf{ECI}_{oc}$ is that it satisfies:

$$\frac{dX_o/X_o}{dX_{o'}/X_{o'}} = \frac{\mathbf{ECI}_{oc}}{\mathbf{ECI}_{o'c}},$$

so the $\mathbf{ECI}_{oc}$ is a useful index of the *relative* impact of the shock on the aggregate exports across countries.

Empirical Implementation for a Given Φ_p Suppose we have a proxy for the size of the price shock in sector p , which remember reduces Chinese export prices to all destinations by a percentage Φ_p . We can then compute an index of the loss of export sales in all destinations other than U.S. and China by implementing the formula above for the case in which $\mathcal{W} = \mathcal{W}_-$, where $\mathcal{W}_-$ is the set of countries in the world except for the U.S. and China.

Notice that because the formula includes exports only, the summation also omits country o 's itself. Relatedly, X_o in the formula is the sum of all of country o 's exports to all destinations other than the U.S. and China. As for χ_{cpd} , in principle this should be computed as the share of total absorption of good p in destination market d accounted for by China, but we shall approximate it by the share of overall imports of good p by country d that originate in China. We do so because we do not have access to information on the domestic sales of producers of good p in d . A way to theoretically rationalize this empirical strategy would be to stipulate preferences as being separable in domestic varieties and imported ones, so that all the relevant CES aggregators sum only over foreign varieties. In such a case, the model would dictate that χ_{cpd} should indeed be computed as the ratio of d 's imports of good p from China over d 's aggregate imports of good p .

Opening the Black Box of Φ_p We next provide a structural interpretation of the reduced-form China export price shock Φ_p introduced in the main analysis. In particular, we show how a U.S. ad-valorem tariff on Chinese imports of a given price p leads to a common decline

in China's export price across non-U.S. destinations when Chinese export supply is upward sloping.

To begin, consider a single product p exported by China to a set of destination countries $d \in \mathcal{W}$, including the United States ($d = U$). China charges a common producer (FOB) price p_{cp} across destinations, while delivered prices are given by

$$\tilde{p}_{cdp} = (1 + \tau_{cdp})p_{cp},$$

where (for simplicity) only the United States imposes a tariff:

$$\tau_{cup} > 0, \quad \tau_{cdp} = 0 \text{ for } d \neq U.$$

Demand in each destination follows the Armington-CES structure described above. As in the rest of the Appendix, we treat destination expenditure on product p , E_{dp} , as locally fixed. Under Armington-CES demand in destination d for product p , China's expenditure share is

$$\chi_{cdp} \equiv \frac{\tilde{p}_{cdp}q_{cdp}}{E_{dp}},$$

and the exact own-price elasticity of demand for the Chinese variety (holding competitor's prices fixed) is

$$\varepsilon_{cdp} \equiv -\frac{d \ln q_{cdp}}{d \ln \tilde{p}_{cdp}} = \sigma_p - (\sigma_p - 1)\chi_{cdp}.$$

We thus have that

$$d \ln q_{cmp} = -\varepsilon_{cdp} d \ln \tilde{p}_{cmp}.$$

Aggregating Chinese export demand across destinations, we have that

$$d \ln Q_{cp} = \sum_{d \in \mathcal{W} - \{c\}} \lambda_{dp} d \ln q_{cdp} = - \sum_{d \in \mathcal{W} - \{c\}} \phi_{cdp} \varepsilon_{cdp} d \ln \tilde{p}_{cmp},$$

where

$$Q_{cp} \equiv \sum_{d \in \mathcal{W} - \{c\}} q_{cdp},$$

and destination quantity shares

$$\phi_{cdp} \equiv \frac{q_{cdp}}{Q_{cp}}, \quad \sum_{d \in \mathcal{W} - \{c\}} \phi_{cdp} = 1.$$

Since $\tilde{p}_{cdp} = (1 + \tau_{cdp})p_{cp}$, we have

$$d \ln \tilde{p}_{cdp} = d \ln p_{cp} + d \ln (1 + \tau_{cdp}).$$

Because only the U.S. tariff changes, we can write

$$d \ln Q_{cp} = -\bar{\varepsilon}_p d \ln p_{cp} - \phi_{cup} \varepsilon_{cup} d \ln (1 + \tau_{cup}),$$

where

$$\bar{\varepsilon}_{cp} \equiv \sum_{d \in \mathcal{W} - \{c\}} \phi_{cdp} \varepsilon_{cdp} \quad \text{and} \quad \varepsilon_{cup} = \sigma_p - (\sigma_p - 1) \chi_{cup}.$$

Assume now that China faces an upward-sloping export supply curve for product p , summarized by

$$d \ln p_{cp} = \frac{1}{\eta_p} d \ln Q_{cp},$$

where $\eta_p > 0$ is the export supply elasticity, which in principle can vary across products.

Substituting the export-demand aggregation into the supply relation and solving yields

$$d \ln p_{cp} = -\phi_{cup} \frac{\varepsilon_{cup}}{\eta_p + \bar{\varepsilon}_{cp}} d \ln (1 + \tau_{cup}). \quad (\text{A.10})$$

How FOB Chinese prices react to the U.S. tariff depend on five key objects: (i) the share of Chinese exports (in quantity terms) that is shipped to the United States, ϕ_{cup} ; (ii) the elasticity of demand faced by Chinese exporters in the U.S., ε_{cup} ; (iii) the elasticity of supply of Chinese exports, η_p ; (iv) a weighted-average elasticity of demand faced by Chinese exporters worldwide, $\bar{\varepsilon}_{cp}$; and (v) the size of the U.S. tariff.

Empirical Implementation To simplify matters, in the main text we implement equation (A.10) for the special case in which the U.S. policy shock is common across sectors ($\tau_{cup} = \tau$) demand parameters σ_p and supply elasticities η_p are common across sectors, and China constitutes a small enough share of absorption in all countries ($\chi_{cdp} \simeq 0$), to approximate the demand elasticity faced by Chinese exporters as given by σ . Under these assumptions, equation (A.10) reduces to

$$d \ln p_{cp} = -\phi_{cup} \frac{\varepsilon}{\eta + \varepsilon} d \ln (1 + \tau),$$

and thus the relative size of the shock in different sectors depends only on $\phi_{cup} = q_{cup}/Q_{cp}$. Finally, if unit values are approximately common across destinations (FOB), ϕ_{cup} is well

approximated by the corresponding export share:

$$\frac{q_{cup}}{Q_{cp}} \simeq \frac{X_{cup}}{\sum_{\mathcal{W}-\{c\}} X_{cdp}} \equiv \Phi_p,$$

which corresponds to equation (6) in the main text.

An Import Competition Index We now consider the impact of the same shock on the domestic producers of a destination market d . As stated in the main text, there is a clear connection between this question and our derivations above, in the sense that we are considering a special case of our shock in which the destination market d under consideration is identical to the origin country o . These considerations have no bearing on the size of the shock Φ_p originating in the U.S., and they are neither relevant, *in theory*, for how the shock impacts market d in general, as proxied by the Chinese absorption share χ_{cpd} . The main difference is that the shock is focused on the domestic market d and that the relevant sales share is the share of domestic sales accounted by sector p , or $s_{dp} = S_{dp} / \sum_{p' \in \mathcal{P}} S_{dp'}$. Overall, the change in domestic sales, referred to as S_d , is given by

$$\frac{dS_d}{S_d} = - \sum_{p \in \mathcal{P}} (\sigma_p - 1) \times \Phi_p \times \chi_{cpd} \times s_{dp}.$$

Ignoring cross-section heterogeneity in demand elasticities ($\sigma_p \simeq \sigma$ for all p), we obtain our import competition index in equation (15) of the main text, or

$$ICI_{dc} = \sum_{p \in \mathcal{P}} \Phi_p \times \chi_{cpd} \times s_{dp},$$

which satisfies

$$\frac{dS_d/S_d}{dS_{d'}/S_{d'}} = \frac{ICI_{dc}}{ICI_{d'c}}.$$

As argued in the main text, when implementing this measure, researchers cannot typically rely on exact measures of s_{dp} , so in our empirical analysis, we proxy s_{dp} with the share of overall exports of country d that are accounted by exports of good p , as indicated in equation (16). For similar reasons, and as in the export competition index, we continue to approximate χ_{cpd} with the share of China in country d 's imports of good p , as in equation (16) in the main text.

An Import Price Decline Index In the main text, we stated that the percentage fall in import prices in a given destination market d can be approximated, up to a first order, with

the following expression

$$\mathbf{IPDI}_{dc} = \sum_{p \in \mathcal{P}} \Phi_p \times \chi_{cdp}^M \times s_{dp}^M,$$

where Φ_p is identical to that in the expressions for $\mathbf{ECI}_{oc}$ and $\mathbf{ICI}_{dc}$ in equations (10) and (15), and where χ_{cdp}^M is the Chinese share in total imports of good p in destination d (which was already our proxy for χ_{cdp} when empirically implementing our export competition index). The term s_{dp}^M in the above equation for $\mathbf{IPDI}_{dc}$ is the share of country d 's imports accounted for by sector p .

We next offer a simple proof of this claim. Define the CES import price index for product p in destination d as

$$P_{dp}^M \equiv \left(\sum_{o \in \mathcal{W} - \{d\}} p_{odp}^{1-\sigma_p} \right)^{\frac{1}{1-\sigma_p}},$$

so that $P_{dp} = \left(p_{ddp}^{1-\sigma_p} + (P_{dp}^M)^{1-\sigma_p} \right)^{1/(1-\sigma_p)}$ is the overall price index for good p in country d .

The associated import expenditure shares are

$$\chi_{odp}^M \equiv \frac{p_{odp}^{1-\sigma_p}}{\sum_{i \in \mathcal{W} - \{d\}} p_{idp}^{1-\sigma_p}}, \quad \sum_{o \in \mathcal{W} - \{d\}} \chi_{odp}^M = 1.$$

A first-order log-differentiation of P_{dp}^M yields the standard Divisia result:

$$d \ln P_{dp}^M = \sum_{o \in \mathcal{W} - \{d\}} \chi_{odp}^M d \ln p_{odp}.$$

Consider now the bilateral shock in (A.3): only China's delivered price changes for product p in market d , so $d \ln p_{cdp} = -\Phi_p$ and $d \ln p_{odp} = 0$ for all $o \neq c$. It follows immediately that

$$d \ln P_{dp}^M = -\chi_{cdp}^M \Phi_p.$$

Next, let the aggregate import price index in destination d be a Cobb–Douglas aggregator over product-level import price indices,

$$P_d^M \equiv \prod_{p \in \mathcal{P}} (P_{dp}^M)^{s_{dp}^M}, \quad \text{with} \quad s_{dp}^M \equiv \frac{M_{dp}}{\sum_{p' \in \mathcal{P}} M_{dp'}},$$

where M_{dp} denotes total imports of product p by destination d .¹⁶ Log-differentiating gives

$$d \ln P_d^M = \sum_{p \in \mathcal{P}} s_{dp}^M d \ln P_{dp}^M.$$

Substituting the product-level result then implies

$$d \ln P_d^M = - \sum_{p \in \mathcal{P}} s_{dp}^M \chi_{cdp}^M \Phi_p.$$

Therefore, the (positive) percentage fall in the aggregate import price index is, up to a first-order approximation,

$$-\frac{dP_d^M}{P_d^M} \approx -d \ln P_d^M = \sum_{p \in \mathcal{P}} \Phi_p \times \chi_{cdp}^M \times s_{dp}^M,$$

which is the expression stated in the text.

Intermediate Inputs We now extend the baseline Armington–CES environment to allow a subset of products, $p \in \mathcal{P}^I \subseteq \mathcal{P}$, to be used as intermediate inputs by downstream producers in each destination market d . For each input product $p \in \mathcal{P}^I$, downstream producers in d assemble a CES composite from origin-specific varieties,

$$Q_{dp}^I \equiv \left(\sum_{o \in \mathcal{W}} q_{odp}^{\frac{\sigma_p - 1}{\sigma_p}} \right)^{\frac{\sigma_p}{\sigma_p - 1}}, \quad \sigma_p > 1,$$

with associated CES price index

$$P_{dp}^I \equiv \left(\sum_{o \in \mathcal{W}} p_{odp}^{1 - \sigma_p} \right)^{\frac{1}{1 - \sigma_p}}.$$

Let E_{dp}^I denote total expenditure by downstream producers in destination d on input p . Conditional on E_{dp}^I , the optimal sourcing shares are identical to those derived above:

$$\chi_{odp}^I \equiv \frac{p_{odp}^{1 - \sigma_p}}{\sum_{i \in \mathcal{W}} p_{idp}^{1 - \sigma_p}}, \quad X_{odp}^I = \chi_{odp}^I E_{dp}^I.$$

Domestic intermediate input sales in sector p are therefore $S_{dp} \equiv X_{ddp}^I = \chi_{ddp}^I E_{dp}^I$.

¹⁶More generally, any smooth aggregator with expenditure weights s_{dp}^M delivers the same first-order log change around the initial allocation.

A Sourcing Competition Index Consider again the bilateral shock in A.3, so that $d \ln p_{cdp} = -\Phi_p$ and $d \ln p_{odp} = 0$ for all $o \neq c$, and take a first-order approximation around an initial allocation holding E_{dp}^I fixed.¹⁷ Since $\chi_{ddp}^I = (p_{ddp}/P_{dp}^I)^{1-\sigma_p}$, we have

$$d \ln S_{dp} = d \ln \chi_{ddp}^I = (\sigma_p - 1) d \ln P_{dp}^I = -(\sigma_p - 1) \chi_{cdp}^I \Phi_p.$$

Aggregating over inputs, let $S_d^I \equiv \sum_{p \in \mathcal{P}^I} S_{dp}$ and define the domestic intermediate-sales weights $s_{dp}^I \equiv S_{dp}/S_d^I$. A first-order log change in aggregate domestic intermediate input sales is then

$$\frac{dS_d^I}{S_d^I} = - \sum_{p \in \mathcal{P}^I} (\sigma_p - 1) \times \Phi_p \times \chi_{cdp}^I \times s_{dp}^I.$$

Ignoring cross-sectional heterogeneity in σ_p and using the fact that good p is either a final good or an input so we can simply write $\chi_{cdp}^I = \chi_{cdp}$, we end up with the sourcing competition index $\mathbf{SCI}_{dc}$ in (??), such that

$$\frac{dS_d^I/S_d^I}{dS_{d'}^I/S_{d'}^I} = \frac{\mathbf{SCI}_{dc}}{\mathbf{SCI}_{d'c}}.$$

Because we do not have data on domestic sales at a disaggregated product level, when empirically implementing the index $\mathbf{SCI}_{dc}$, we proxy the domestic share s_{dp}^I with the share of overall intermediate input exports of country d that are accounted by exports of input p , as in equation (16) in the main text.

A Sourcing Price Decline Index Next, define the CES import price index for intermediate input $p \in \mathcal{P}^I$ in destination d as

$$P_{dp}^{M,I} \equiv \left(\sum_{o \in \mathcal{W} \setminus \{d\}} p_{odp}^{1-\sigma_p} \right)^{\frac{1}{1-\sigma_p}}.$$

Under the same China-only shock, $d \ln P_{dp}^{M,I} = -\chi_{cdp}^{M,I} \Phi_p$, where $\chi_{cdp}^{M,I}$ is China's expenditure share in destination d 's imports of input p . Aggregating across intermediate inputs with a Cobb–Douglas index, and assuming no cross-sectional variation in demand elasticities,

$$P_d^{M,I} \equiv \prod_{p \in \mathcal{P}^I} \left(P_{dp}^{M,I} \right)^{s_{dp}^{M,I}}, \quad s_{dp}^{M,I} \equiv \frac{M_{dp}^I}{\sum_{p' \in \mathcal{P}^{input}} M_{dp'}^I},$$

¹⁷This “local” closure obtains exactly, for example, if downstream production uses a Cobb–Douglas aggregator over intermediate inputs so that expenditures on each input are locally proportional to downstream scale.

implies the first-order result

$$-d \ln P_d^{M,I} = \sum_{p \in \mathcal{P}^I} \Phi_p \times \chi_{cdp}^{M,I} \times s_{dp}^{M,I}.$$

Because a good p is either a final good or an input so we can simply write $\chi_{cdp}^{M,I} = \chi_{cdp}^M$. We then have that for our empirical proxy of Φ_p , we have the following sourcing price decline index

$$SPDI_{dc} = \sum_{p \in \mathcal{P}^I} \Phi_p \times \chi_{cdp}^M \times s_{dp}^{M,I},$$

which corresponds to equation (20) in the main text.

A Rerouting Potential Index We next construct an index that captures the extent to which country i 's exports to the United States (u) increase in response to a reduction in the price of intermediate inputs imported from China. Unlike the export competition index, which operates through a demand-side shock in third markets, here the mechanism is a supply-side cost shock: a fall in the price of Chinese intermediate inputs lowers marginal costs in i and thus reduces the price of the varieties produced by i .

To allow for product-level heterogeneity in the marginal-cost reduction, let firms producing output good p in country i use a Cobb–Douglas bundle of intermediate inputs $z \in \mathcal{P}^I$ with cost shares a_{pz} , so that the relevant intermediate input price index is

$$P_{ip}^I \equiv \prod_{z \in \mathcal{P}^I} \left(P_{iz}^{M,I} \right)^{a_{pz}}, \quad \sum_{z \in \mathcal{P}^I} a_{pz} = 1,$$

where $P_{iz}^{M,I}$ is the CES import price index for intermediate input z in country i as defined above. Under the same China-only shock, we have $d \ln P_{iz}^{M,I} = -\chi_{ciz}^M \Phi_z$, where χ_{ciz}^M is China's import expenditure share in country i for input z . It follows that the first-order change in the intermediate input price index relevant for producing output p is

$$-d \ln P_{ip}^I = \sum_{z \in \mathcal{P}^I} a_{pz} \times \Phi_z \times \chi_{ciz}^M.$$

Assuming that this reduction in input costs translates one-for-one into a reduction in the producer price of the i -variety of product p , Armington CES demand in the U.S. implies that, holding the U.S. product-level price index fixed,

$$\frac{dX_{iup}}{X_{iup}} = -(\sigma_p - 1) d \ln P_{ip}^I.$$

Aggregating across products with baseline export shares to the U.S., $s_{iup} \equiv X_{iup}/X_{iu}$, we obtain

$$\frac{dX_{iu}}{X_{iu}} = - \sum_{p \in \mathcal{P}} s_{iup} \times (\sigma_p - 1) \times d \ln P_{ip}^I.$$

If the elasticity of demand does not vary across products, then notice that the rerouting potential index can be written as

$$\mathbf{RPI}_{ciu} \equiv \sum_{p \in \mathcal{P}} s_{iup} \left(\sum_{z \in \mathcal{P}^I} a_{pz} \times \Phi_z \times \chi_{ciz}^M \right). \quad (\text{A.11})$$

If demand elasticities and pass-through are common across products and countries, then $\mathbf{RPI}_{ciu}$ is proportional to the predicted percentage change in country i 's exports to the U.S., and it satisfies

$$\frac{dX_{iu}/X_{iu}}{dX_{i'u}/X_{i'u}} = \frac{\mathbf{RPI}_{ciu}}{\mathbf{RPI}_{ci'u}}.$$

Empirical implementation of the RPI without an Input–Output table In principle, one would ideally discipline the coefficients a_{pz} using a product-level input–output (IO) table that maps output goods into their intermediate input requirements. In the absence of such data (or when the IO table cannot be consistently harmonized with HS trade classifications), we propose a set of trade-based approximations that impose structure on a_{pz} using HS nesting. Specifically, for $\ell \in \{6, 4, 2\}$, define the HS- ℓ “neighborhood” of output good p as

$$\mathcal{P}^{I,(\ell)}(p) \equiv \{z \in \mathcal{P}^I : HS_\ell(z) = HS_\ell(p)\},$$

where $HS_6(p)$ is the full HS-6 code (so that $\mathcal{P}^{I,(6)}(p)$ selects only $z = p$), while $HS_4(p)$ and $HS_2(p)$ denote the HS-4 and HS-2 aggregations of p .

We then proxy the (unknown) IO weights a_{pz} by country i 's baseline intermediate import shares within the neighborhood of p :

$$s_{iz|p}^{M,I,(\ell)} \equiv \frac{M_{iz}^I}{\sum_{z' \in \mathcal{P}^{I,(\ell)}(p)} M_{iz'}^I}, \quad z \in \mathcal{P}^{I,(\ell)}(p).$$

This yields a product-specific sourcing price decline index (SPDI) at level ℓ :

$$\mathbf{SPDI}_{icp}^{(\ell)} \equiv \sum_{z \in \mathcal{P}^{I,(\ell)}(p)} \Phi_z \times \chi_{ciz}^M \times s_{iz|p}^{M,I,(\ell)}.$$

The three values of ℓ correspond to progressively broader assumptions about input linkages: HS-6 restricts inputs to the same HS-6 code as the output (an extreme “own-input” assumption),

HS-4 allows inputs within the same HS-4 heading, and HS-2 allows inputs within the same HS-2 chapter. These alternatives can be interpreted as a robustness set that brackets plausible input–output linkages when the true a_{pz} matrix is unavailable.

Replacing the IO-based object in (A.11) with these proxies gives the empirically implementable rerouting potential index

$$RPI_{ciu}^{(\ell)} \equiv \sum_{p \in \mathcal{P}} s_{iup} \times SPDI_{icp}^{(\ell)}.$$

Finally, notice that if one instead assumes a common economy-wide intermediate input bundle across all outputs (i.e., a_{pz} does not vary with p and is proportional to $s_{iz}^{M,I}$), then $SPDI_{icp}^{(\ell)}$ collapses to the aggregate $SPDI_{ic}$ defined above, and product-level composition becomes irrelevant for the predicted percentage change in X_{iu} .

B Empirical Constructions

Trade Data. Trade data comes from the Trade Data Monitor (TDM), which contains monthly imports and exports at the six-digit commodity level. Monthly observations are collapsed to annual totals. Country codes are standardized into ISO 3-letter codes. We substitute missing flows when possible with mirrored observations from other countries. For example, if China does not report any imports from the US, but the US reports exports to China, we use the US export data to fill in the missing Chinese imports. After mirroring, we default to using import data as the measure of trade flows. For earlier trade data, namely data from 2017 used to construct trade-weighted average elasticities and earlier baseline shares for index robustness, we use [Gaulier and Zignago \(2010\)](#).

Elasticities. Elasticity estimates come from [Soderbery \(2018\)](#). Demand elasticities σ_{dp} are reported at the destination-HS four-digit level. Inverse-supply elasticities η_{odp} are reported as the origin-destination-HS four-digit level. Elasticities are cross-walked from their 1992 vintage to the 2017 vintage used in the trade data. When reporting demand elasticities, the author truncated estimates to 131.05, we opt to do the same to our working estimates. In addition, we apply the same truncation to supply elasticities.¹⁸ We take the average of inverse-supply across destinations using export values in 2017, to get an elasticity, η_{op} , that is at the origin-HS4 level. Using an earlier period avoids endogeneity between the trade weights and tariff shocks being studied.

¹⁸[Soderbery \(2018\)](#) drops supply elasticity in the top and bottom 0.5% when reporting estimates, but this still keeps many supply elasticities in the order of magnitude of 10^6 . We therefore opt to truncate elasticities.

In addition, [Soderbery \(2018\)](#) reports standard errors for estimated elasticities. Let θ denote an estimated elasticity, and σ_θ the associated standard error. We drop any elasticities where $\theta/\sigma_\theta < 1.96$. There are estimates which can not be distinguished from zero under standard significance conventions.

Elasticities are reported with the 1992 HS vintage, we cross-walk there to 2017 HS vintage using the Concordance R package ([Liao et al., 2020](#)). Note that the [Soderbery \(2018\)](#) elasticities are estimated from trade data over the period 1993-2007, and there is certainty scope for preferences and technology to change before our baseline period of 2023. Hence, we continue to report "basic" measures as they are free of any structure elasticities and simply use observed trade shares.

Tariffs. Tariff data comes from [Rodríguez-Clare et al. \(2025\)](#). This data primarily contains tariff data for the US. We consider two tariff windows. First, a short difference surrounding the second Trump administration: January 2025 to September 2025 (the final period of available data). Second, a long tariff difference beginning from before the first Trump administration's tariff policies to present, January 2018 to September 2025.

Additional Data. The SGI requires input-output data at the HS six-digit level. We use binary input-output connections provided by [Fetzer et al. \(2024\)](#). One limitation of this data is that no data on the extent of input usage is provided, outside of the binary classification of whether an upstream HS code is used in the production of another downstream HS code. Therefore, we give a uniform weight to each input.

The Upstream and Downstream variants of the CGI require classifying goods as inputs or final goods. To do so we use the upstreamness measure from [Antràs et al. \(2012\)](#), and classify any goods with an upstreamness greater than 2 as intermediate goods and less than 2 as final goods. The upstreamness measure is reported for Input-Output Commodity Codes which we crosswalk to HS codes with data provided by the BEA.

B.1 Data Coverage

In Table [B.1](#) we report the fraction of products for which we observe the relevant elasticities needed to compute the shock to Chinese export prices [\(7\)](#). At the very least, these products are included in our analysis. For computing subsequent measures, we may need to drop additional trade flows because of additional missing parameters. For example, the ECI requires import demand elasticities at all export markets, should these be absent a trade flow needs to be excluded from the ECI construction. In Table [B.2](#) we report the fraction of trade flows where we observe the relevant price shock and import-demand elasticity.

Another consequence of partial coverage is then when computing indices that take an average over products, e.g. the ECI, we must re-normalize export shares so that they sum to one over the set of products for which we have the relevant data. Otherwise, indices will be correlated with coverage statistics. Since some products need to be dropped, this leads to the possibility that the indices do not accurately reflect

Table B.1: Product Coverage

Component	HS6 products	% products	% trade value
US-China demand elasticity (ε_{cup})	6,303	71.8%	85.2%
Chinese supply elasticity (η_{cp})	6,611	75.3%	79.9%
Tariff change ($\Delta \log(1 + \tau)$)	5,232	59.6%	89.6%
All components	3,852	43.9%	69.2%
Upstreamness measure	4,685	53.4%	71.2%

Note. For each 'component' row, 'HS6 products' reports the number of HS codes where we observe the component, '% products' the share of products we observe the component, and '% trade value' the share of trade flows accounted for by the observed products. 'All components' refers to HS codes that contain demand elasticity, supply elasticities, and tariff changes needed to compute Chinese price shocks.

Table B.2: Trade Flow Coverage

Component	% trade flows	% trade value
Price shock (Φ_p)	78.6%	69.2%
Demand elasticity (σ_{dp})	82.8%	83.7%
All components	68.8%	65.3%

Note. For each 'component' row, '% trade flows' reports the fraction of trade flows where we observe the component and '% trade value' the share of trade flows accounted for by the observed components.

C Additional Figures and Tables

Table C.1: Price Shocks

Rank	HS6	Description	ϕ_{cup}	σ_{up}	ε_{cup}	η_{cp}	$\bar{\varepsilon}_{cp}$	$\Delta\tilde{\tau}_{cup}$	$d \log p_{cp}$	$d \log p_{cp}^{Long}$
1	850790	Electric accumulators; parts n.e.c. in headi...	0.472	2.6	2.1	0.7787	1.844	0.793	0.295	0.093
2	842549	Jacks and hoists; (other than hydraulic), of...	0.411	2.9	1.9	0.8862	1.946	0.824	0.231	0.352
3	940290	Furniture; for medical, surgical, veterinary...	0.350	78.2	50.2	1.5914	21.261	0.262	0.202	0.213
4	732181	Domestic appliances; non-electric, (other th...	0.567	3.1	1.5	0.6258	1.551	0.510	0.201	0.212
5	830230	Mountings, fittings and similar articles; fo...	0.322	2.6	2.3	0.6315	2.481	0.813	0.193	0.399
6	732119	Cooking appliances and plate warmers; for so...	0.594	3.1	1.2	0.6258	1.315	0.531	0.189	0.163
7	901510	Rangefinders	0.248	2.3	2.0	0.3709	1.902	0.824	0.180	0.374
8	293080	Organo-sulphur compounds; aldicarb (ISO), ca...	0.943	2.7	1.0	0.4246	1.006	0.248	0.164	0.056
9	940210	Chairs; dentists', barbers' or similar chair...	0.237	78.2	30.0	1.5914	9.752	0.262	0.164	0.173
10	090112	Coffee; decaffeinated, not roasted	0.861	3.8	3.8	1.3189	3.782	0.254	0.163	0.209
11	400932	Rubber; vulcanised (other than hard rubber),...	0.409	2.7	2.5	0.8919	2.737	0.557	0.155	0.297
12	903149	Optical instruments and appliances; for meas...	0.073	131.1	120.9	0.6481	11.964	0.215	0.151	0.231
13	732111	Cooking appliances and plate warmers; for ga...	0.400	3.1	2.1	0.6258	1.870	0.439	0.150	0.248
14	903180	Instruments, appliances and machines; for me...	0.078	131.1	120.8	0.6481	13.593	0.215	0.142	0.301
15	851130	Ignition or starting equipment; distributors...	0.274	3.0	2.4	1.8326	1.999	0.813	0.137	0.220
16	030615	Crustaceans; frozen, Norway lobsters (Nephro...	0.839	5.6	5.0	1.9655	4.690	0.215	0.135	0.017
17	420100	Saddlery and harness; for any animal (includ...	0.252	30.1	18.2	0.6042	6.565	0.211	0.135	0.306
18	300449	Medicaments; containing alkaloids or their d...	0.804	4.2	4.0	0.5607	3.873	0.182	0.133	0.080
19	030563	Fish; salted or in brine, not dried or smoke...	0.949	2.8	2.6	1.3304	2.588	0.213	0.133	0.000
20	732189	Domestic appliances; non-electric, (other th...	0.395	3.1	1.7	0.6258	1.954	0.501	0.130	0.171

Note. The term ϕ_{cup} is the share of Chinese exports of goods p going to the US, σ_{up} is the US import demand elasticity for good p , $\varepsilon_{cup} = \sigma_{dp} - (\sigma_{dp} - 1) \times \chi_{cup}$ where χ_{cup} is the share of US import expenditure in good p on goods coming from China, $\bar{\varepsilon}_{cp}$ is the export-weighted average ε_{cup} faced by China across export markets, $\Delta\tilde{\tau}_{cup} \equiv \Delta \log(1 + \tau_{cup})$ is the change in tariffs between January 2025 to September 2025, and the price shock is $d \log p_{cp} = \Delta\tilde{\tau}_{cup} \times \phi_{cup} \times \varepsilon_{cup} / (\bar{\varepsilon}_{cp} + \eta_p)$. In the baseline we use trade data from 2023 to compute shares. For long-term price shocks, $d \log p_{cp}^{Long}$, we use trade data from 2017 to compute shares and use the tariff difference between January 2018 and September 2025.

Table C.2: Across-Index Rank Correlation

	ECI	ESI	ICI	ISI	CGI	SGI
ECI	1.000	—	—	—	—	—
ESI	0.675	1.000	—	—	—	—
ICI	0.650	0.414	1.000	—	—	—
ISI	0.235	0.635	0.121	1.000	—	—
CGI	-0.177	-0.159	0.169	0.085	1.000	—
SGI	-0.191	-0.100	0.035	-0.022	0.451	1.000

Note. Each cell reports a Spearman’s rank correlation coefficient between the country-specific indices referenced in the row and column.

Table C.3: Within-Index Rank Correlation

Index	Full vs. Basic	Full vs. Long	Full vs. Basic Full
ICI	0.900	0.968	0.611
ECI	0.957	0.977	0.895
CGI	0.988	0.967	0.881
SGI	0.877	0.970	0.727

Note. For each index (labeled in the row), each cell reports a Spearman’s rank correlation coefficient between the variants (labeled in the column) of the country-specific indices. ‘Full’ is the baseline index, ‘Basic’ is to the index without any trade elasticities or tariffs but the same set of products as ‘Full’, ‘Long’ is the baseline index but with tariff differences over the 2018-2025 period, and ‘Basic Full’ is the same as ‘Basic’ but extended to include the full set of products. Product coverage differs between ‘Full’ and ‘Basic Full’ because of missing elasticity estimates as discussed in Table B.1.